

IgA Nephropathy and IgA Vasculitis

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CHAPTER OUTLINE

IgA NEPHROPATHY IN ADULTS, 980
IgAN IN CHILDREN, 996

IgA VASCULITIS IN ADULTS, 1001
IgA VASCULITIS IN CHILDREN, 1006

KEY POINTS

- IgA nephropathy is the most common primary glomerular disease worldwide.
- IgA nephropathy carries a high life-time risk of kidney failure.
- Production of poorly O-glycosylated polymeric IgA1 at mucosal sites underlies IgAN pathogenesis.
- Treatment of IgAN should address both generic drivers of nephron loss and IgA-mediated inflammatory pathways.
- IgA vasculitis has a higher incidence in children, but more frequent kidney involvement is seen in adults.

IgA NEPHROPATHY IN ADULTS

INTRODUCTION

The novel application of immunofluorescence microscopy to kidney tissue by Jean Berger, supported by the electron microscopy findings of Nicole Hinglais, led to the first description of glomerulonephritis with mesangial IgA deposition in 1968.¹ By 1975, this disease entity was recognized internationally as Berger disease² and is now referred to as IgA nephropathy (IgAN). It is the most common primary cause of glomerular disease worldwide.³

EPIDEMIOLOGY

The epidemiology of IgAN varies significantly across geographic regions. It is estimated to account for 40%, 22%, and 12% of all glomerular disease diagnoses in Asia, Europe, and North America, respectively.⁴ Exact incidence rates are difficult to calculate: Most kidney biopsy series lack data on the size of the referral population, and national kidney biopsy registries are few. However, estimates from Europe include an annual incidence of 2.8/100,000 in Western France,⁵ 0.84/100,000 in Italy,⁶ 1.9/100,000 in Holland,⁷ and 1.1/100,000 in the Czech Republic.⁸ In Japan, the estimated annual incidence is higher at 3.9 to

4.5/100,000,^{8,9} whereas in Southern California it is lower at 0.7/100,000.¹⁰ Local pockets of disease have been identified in Zuni Native American Indians in the southern United States and among Australian First Nations people.¹¹

The point prevalence of IgAN in Europe estimated from national registries has been estimated as 2.53 per 10,000 people.¹² To account for IgAN in patients who have not undergone kidney biopsy, a genetic risk score based on known risk polymorphisms was applied to a U.K. Biobank, and it was estimated that about 20% of cases with unspecified hematuria probably have underlying IgAN.¹³ Because IgAN is often suspected by performing urinalysis and can only be diagnosed on kidney biopsy, the incidence of IgAN is influenced by local public health policies regarding urinalysis and kidney biopsy practice. For example, urinalysis is currently performed in school-age children as part of national kidney disease screening programs in Japan, South Korea, and Taiwan,¹⁴ enabling detection of IgAN that might not otherwise have come to clinical attention. The threshold to proceed to a kidney biopsy to evaluate isolated microscopic hematuria is also lower in these countries. Nevertheless, large-scale genome-wide association (GWA) studies suggest that persons of Asian ancestry have a higher prevalence of IgAN disease susceptibility loci, genetically predisposing them to develop IgAN. In contrast, persons of recent African ancestry have a much lower genetic risk.¹⁵

This finding is supported by kidney biopsy series from Africa showing a low frequency of IgAN across the continent,¹⁶ as well as from U.S. data indicating a lower frequency of IgAN among African-Americans.¹⁷ The higher incidence of IgAN in Asian-Pacific ethnic groups is also accompanied by a more rapid disease trajectory, even after migration.¹⁸ Therefore the higher reported incidence of IgAN in Asian countries is likely due to a combination of biologic and health systems factors, with more prominent diseases in these regions further influencing more proactive case detection. For more on IgAN in the Far East, see [Chapter 78](#).

A male predominance is reported in Europe, where men are affected up to threefold more frequently than women. In East Asia, men and women are more equally affected.^{19,20} Mean age at diagnosis is during the fourth decade of life across European, American, and Asian populations,²¹ although age at diagnosis appears to be increasing, possibly reflecting a higher rate of detection in older adults rather than a change in the natural history of the disease.²²

PATHOGENESIS

IMMUNOGLOBULIN A

In human serum, immunoglobulin A (IgA) has a concentration of 1 to 3 mg/mL, making it the second most concentrated immunoglobulin in serum after IgG.²³ However, when mucosal secretions are considered, IgA is by far the most abundantly produced immunoglobulin in the body. It is estimated that ~21 mg of IgA per kg are secreted into the circulation per day, while 45 mg of IgA per kg are secreted onto mucosal surfaces per day, giving a total production of 66 mg/kg of IgA per day²⁴—more than all of the other immunoglobulins combined.²⁵

In humans, there are two subclasses of IgA: IgA1 and IgA2.^{26,27} IgA1 represents about 90% of steady-state serum IgA while the other 10% is made up of IgA2. The ratio of IgA1 to IgA2 in colostrum is 65:35, in saliva 60:40, in jejunal fluid 70:30, and in the colon 35:65.²⁵ Serum IgA originates primarily from bone marrow antibody-secreting cells (ASCs) and is 85% to 90% monomeric, while almost all (>95%) of IgA secreted into the gastrointestinal (GI) tract is in polymeric form. Both IgA1 and IgA2 subclasses are capable of forming dimeric and larger, polymeric, molecules (pIgA). IgA molecules undergo considerable posttranslational modification, especially glycosylation, which accounts for up to 10% of their molecular weight. IgA1 can undergo both *N*- and *O*-glycosylation, whereas IgA2 is only *N*-glycosylated.

O-glycosylation of the IgA1 molecule is restricted to the serine (Ser) and threonine (Thr) residues in the hinge region, which lies between the CH1 and CH2 domains of the α 1 heavy chain. At any time, only three to six of the nine threonine and serine residues with potential for *O*-glycosylation in the IgA1 hinge region are occupied by glycans. The three core *O*-glycosylated hinge-region residues are Thr₂₂₈, Ser₂₃₀, and Ser₂₃₂. Thr₂₂₅ and Thr₂₃₆ are occasionally glycosylated, while Thr₂₃₃ is rarely glycosylated.²⁸⁻³⁰ The process of *O*-glycosylation of the IgA1 hinge region has been well characterized. Glycosylation, in contrast to glycation, is an enzyme-driven process. The addition of GalNAc to a threonine/serine residue is catalyzed by UDP-N-acetylgalactosaminyltransferase-2.³¹ Sequential

addition of galactose, to form the Core-1 structure, is mediated by β 1-3 galactosyltransferase (C1 β 3GalT1), which transfers galactose from a UDP-galactose donor; this process is facilitated by a chaperone, Core-1- β 3-Gal-T specific molecular chaperone (COSMC).³² This Core-1 structure can then undergo sialylation by α 2-6 linkage to GalNAc, mediated by the sialyltransferase ST6GalNAcII and/or additional α 2-3 sialylation of the galactose residue by a separate sialyltransferase. Sialic acid generally functions as a capping residue on *O*-glycan chains, so alternatively, if α 2-6 sialylation of *O*-GalNAc occurs before transfer of galactose to the founding *O*-GalNAc residue, the process terminates at that point.^{33,34}

PATHOPHYSIOLOGY

Current understanding of IgAN pathogenesis follows the widely accepted “four-hit hypothesis,” first proposed by Suzuki and colleagues,³⁵ which comprises increased generation of deglycosylated IgA1 (dg-IgA1) (Hit 1), production of autoreactive antibodies targeting dg-IgA1 (Hit 2), leading to formation of immune complexes (Hit 3), which deposit in the glomerular mesangium, triggering an inflammatory cascade leading to kidney damage (Hit 4). ([Fig. 33.1](#))

Hit 1: Increased Generation of De-Glycosylated IgA1

Changes in the circulating pool of IgA1 *O*-glycoforms in IgAN (collectively known as deglycosylated IgA1, dg-IgA1) were first reported using lectin-based assays, whereby serum IgA1 from IgAN patients had reduced affinity for Jacalin lectin.^{36,37} Later studies, using mass-spectrometry-based techniques, confirmed decreased sialylation and galactosylation of *O*-glycans in the hinge region of deposited IgA1.^{38,39} Interestingly, reduced sialylation emerged as the trait most strongly associated with lectin measurements, IgAN diagnosis, and decline in renal function.⁴⁰ Since the availability of the KM55 monoclonal antibody, there have been multiple reports of increased serum concentrations of dg-IgA1 in populations of patients with IgAN from across the world.⁴¹⁻⁴³ Importantly, these changes in the circulating IgA1 *O*-glycoform pool in IgAN are reflected in the kidney. Enrichment of differentially glycosylated IgA1 within the mesangial IgA deposits was first suggested by Monteiro and colleagues⁴⁴ on the basis of the altered charge of IgA1 in glomerular eluates. Confirmation of enhanced deposition of poorly galactosylated IgA1 *O*-glycoforms in the kidneys was later confirmed in two studies, one from the UK and one from Japan.^{45,46}

The source of these pathogenic IgA1 glycoforms has been an area of intense research over the past 40 years. Initial studies showing the polymeric nature of mesangial IgA1 pointed to a mucosal origin of deposited IgA.^{44,47-49} But bone marrow aspirates from IgAN display increased numbers of plasma cells producing polymeric IgA1 compared with healthy subjects,^{50,51} while duodenal plasma cells expressing polymeric IgA1 were decreased in IgAN patients.⁵² These findings gave investigators reason to believe that pathogenic pIgA could arise from the systemic, rather than mucosal, immune system.

However, in a mouse model, Emancipator and colleagues⁵³ showed that oral immunization triggers mesangial IgA deposition, resembling human IgAN,

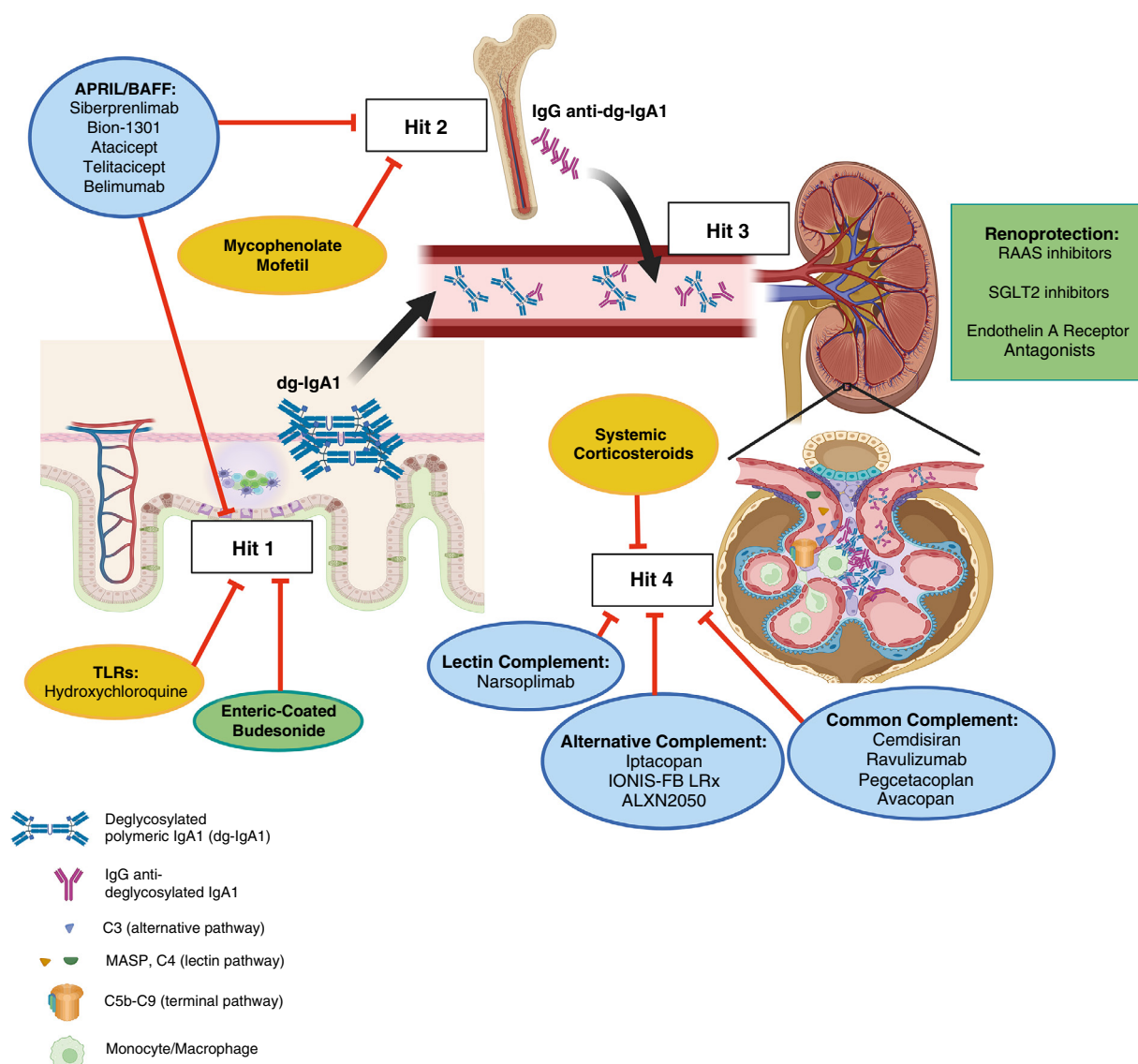


Fig. 33.1 Four-hit hypothesis for pathophysiology of IgAN and drug targets. (From Gleeson PJ, O'Shaughnessy MM, Barratt J. IgA nephropathy in adults—treatment standard. *Nephrol Dial Transplant*. 2023;38(11):2464–2473.)

with accompanying immunogen deposition. Barratt and colleagues⁵⁴ demonstrated that patients with IgAN have an exaggerated polymeric IgA response to *Helicobacter pylori* infection, consistent with an intestinal mucosal origin of pathogenic IgA. Building on this work, Smith and colleagues^{55,56} later showed that the exaggerated pIgA response to *H. pylori* seen in IgAN was enriched for poorly galactosylated IgA1 O-glycoforms, consistent with a mucosal origin of dg-IgA1. Levels of secretory IgA (SIgA), which is exclusively produced at mucosal surfaces, are elevated in the serum patients with IgAN, and mesangial deposition of SIgA has been reported in 15% to 30% of cases. In one case, relative to serum concentrations, the accumulation of SIgA in glomeruli was 120-fold that of IgA.⁵⁵

It has been suggested that during physiologic trafficking of mucosally primed lymphocytes back to the mucosa, there is mishoming of some of these cells to systemic sites such as the bone marrow in IgAN, where they secrete “mucosal” pIgA directly into the circulation. Indeed, changes in the function of circulating peripheral B cells have been reported

by multiple investigators. Suzuki and colleagues showed that circulating B cells from IgAN patients secrete more polymeric galactose-deficient IgA1 than healthy subjects, while Qin and colleagues⁵⁷ reported that peripheral B lymphocytes in IgAN had lower baseline levels of COSMC and were therefore more likely to synthesize galactose-deficient IgA1. It has also been shown that factors within the B cells' immediate microenvironment are capable of modifying IgA1 O-glycosylation.

IL-6, and to a lesser extent IL-4, reduces expression of C1β3GalT1 and increases expression of ST6GalNacII, resulting in reduced galactosylation of IgA1 hinge-region O-glycans.⁵⁸ Activation of numerous pattern recognition receptors (PRRs) including Toll-like receptor 9 (TLR9) has also been shown to modify glycosyltransferase expression and IgA1 O-glycosylation.

Increased levels of dg-IgA1 alone, however, are insufficient to cause IgAN. Serum levels of dg-IgA1 are similarly elevated in healthy first-degree relatives of patients with IgAN.⁵⁹ Gharavi and colleagues^{60–62} demonstrated that the

serum level of dg-IgA1 is an inheritable trait, and separate quantitative trait GWA studies have identified SNPs associated with *C1GALT1*, which encodes C1β3GalT1, and *COSMC* as being significantly associated with serum dg-IgA1 levels after addition of exogenous sialidase. Interestingly, *C1GALT1* and *COSMC* have not been identified as risk loci for IgAN in any published GWA studies. Single nucleotide polymorphisms (SNPs) in the genes encoding APRIL (*TNFSF13*) and one of its receptors, Transmembrane Activator and CAML Interactor (TACI), are associated with an increased risk of developing IgAN.⁶³ New data show that mucin-degrading bacterial species within the gut microbiota, which are found

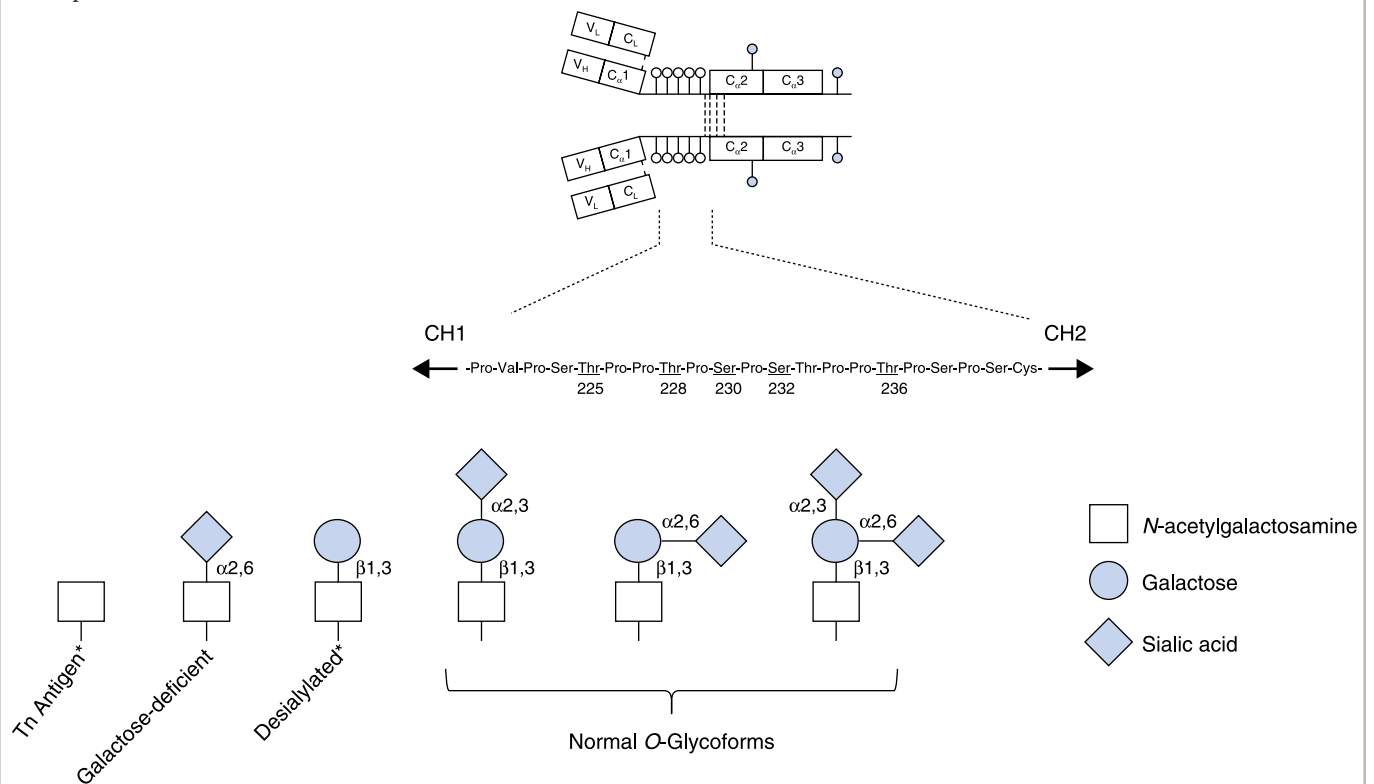
in higher levels in patients with IgAN, have the ability to deglycosylate IgA1 in the intestinal lumen. Generated dg-IgA1 can then pass back across the intestinal barrier and into the circulation by retro-transcytosis.⁶⁴

Taken together, these data support the mucosal immune system as the predominant source of pathogenic dg-IgA1 in IgAN, where the prevailing microenvironment favors generation of IgA1 with reduced hinge-region *O*-glycosylation, influenced by the effect of the local microbiome, microbial-derived pathogen-associated molecular pattern (PAMP) activation of PRRs, and two key pro-IgA B cell survival factors: BAFF and A Proliferation Inducing Ligand (APRIL).^{64,65}

Terminology Regarding Glycosylation of IgA1

It has been known for years that glycosylation of IgA1 deposited in the renal mesangium of patients with IgA nephropathy is altered. Human glycoproteins undergo *N*-glycosylation (glycans attached to asparagine amino acid residues) and some, including IgA1, also undergo *O*-glycosylation (glycans attached to threonine or serine amino acid residues) through a process called posttranslational modification. While *N*-glycosylation of IgA1 has also been shown to be different in IgAN, the focus has been on *O*-glycosylation because it is altered *O*-glycosylation in the threonine- and serine-rich hinge region of IgA1 that is the target of autoantibodies in this autoimmune disease. Recognition of deglycosylated *O*-glycans in the IgA1 hinge region by autoantibodies leads to the formation of large nephritogenic autoimmune complexes.

O-glycans are made up of saccharide residues including GalNAC, galactose, and sialic acid. The pathogenic *O*-glycoforms in the hinge region of IgA1 found in IgAN lack galactose, which has led to them being referred to as “galactose deficient” or “hypogalactosylated.” However, they also lack sialic acid residues. Laboratory techniques used to investigate the origins of these pathogenic IgA1 *O*-glycoforms have often added an extrinsic neuraminidase to reproduce the pathogenic glycoforms found in IgAN. Neuraminidases remove sialic acid residues from *O*-glycans. If no galactose residues are present, then the underlying GalNAC residue will be exposed, as is the case in patients with IgAN. However, the pathogenic *O*-glycoform of IgA1 in IgAN naturally lacks both sialic acid and galactose. As such, we use the term “deglycosylated” IgA1 (dg-IgA1) to account for the lack of both saccharides when referring to the pathogenic glycoform of IgA1 found in patients.



*Indicates the two pathogenic glycoforms.

Hit 2: Production of Autoreactive Antibodies Targeting Dg-IgA1

As an increase in serum dg-IgA1 levels is not sufficient to cause IgAN, additional disease triggers must be present and the “second hit” is proposed to be the presence of IgA and IgG antibodies reactive against dg-IgA1, resulting in the formation of circulating IgA immune complexes. Tomana and colleagues demonstrated the presence of serum IgA and IgG antibodies specific for dg-IgA1 in IgAN. These IgG anti-dg-IgA1 antibodies are predominantly of the IgG2 isotype, are elevated in IgAN, display specificity for IgA1 O-glycoforms displaying the Tn (GalNAc) antigen,^{66-68,69} and can be identified in glomerular eluates from IgAN kidney biopsies.⁷⁰ The Tn antigen is defined as the monosaccharide structure N-acetylgalactosamine (GalNAc) linked to serine or threonine by a glycosidic bond, and it is recognized as an antigen that is reported to exist broadly on the surfaces of bacteria and viruses. It has been suggested that the dg-IgA1 specific antibodies reported in IgAN are actually physiologic antimicrobial antibodies that cross-react with human gd-IgA1 through a process known as molecular mimicry.^{71,72}

Hit 3: Formation of Gd-IgA1 Containing Immune Complexes

The precise mechanisms that lead to IgA immune complex formation in IgAN are incompletely understood, with formation likely occurring both in the circulation and in situ in the mesangium.³⁵ Self-aggregation of dg-IgA1, interaction of dg-IgA1 with IgA and IgG autoantibodies, and interaction of dg-IgA1 with a range of other serum proteins, including complement proteins, α 1-microglobulin, and soluble CD89, have all been proposed as key drivers of pathogenic IgA immune-complex formation.^{66-68,73,74} The Fc fragment of IgA receptor (FCAR) is a human gene that encodes the transmembrane receptor Fc α RI, also known as CD89 (Cluster of Differentiation 89). Fc α RI specifically binds to the heavy-chain constant region of IgA antibodies, and its extracellular domain can be shed to exist in a soluble form either freely or bound to IgA. IgA immune complexes are normally cleared from the circulation through asialoglycoprotein receptor (ASGPR)-mediated endocytosis in the liver. It has been hypothesized that the immune complexes generated in IgAN are too large to enter the space of Disse in the liver and cannot therefore engage with the hepatocyte ASGPR, leading to their persistence in the circulation. By contrast, these large immune complexes can cross the fenestrations of the glomerular capillary endothelium, but not the GBM, and therefore become trapped in the mesangium.⁷⁵

Hit 4: Deposition of Immune Complexes in the Glomerular Mesangium Causing Inflammation

The principal cell that reacts to glomerular IgA accumulation is the mesangial cell. Deposited IgA is recognized by the mesangial cell transferrin receptor (CD71), which is upregulated in IgAN and preferentially binds IgA1 O-glycoforms with lower levels of sialylation and galactosylation.⁷⁶ IgA-induced activation of mesangial cells triggers cell proliferation and release of proinflammatory and profibrotic mediators including IL-6, IL-8, IL-1 β , TNF- α , IFN- γ inducible protein 10, monocyte chemotactic peptide-1, and TGF- β .⁷⁷⁻⁸² These mediators act locally in the mesangium and also cross the GBM, resulting in direct effects on podocytes and tubular epithelial cells. These

deleterious signals are amplified by concomitant activation of the complement system by IgA immune complexes in the mesangium. Specific IgA1 O-glycoforms enriched in mesangial deposits promote activation of both the alternative pathway (AP) and the lectin pathway (LP).^{83,84} IgG anti-dg-IgA1 antibodies are predominantly of the IgG2 isotype, which is less efficient at activating complement, and IgG2 has a much lower affinity for the Fc receptor. Therefore classical pathway activation is rarely seen in IgAN due to the paucity of IgG1 and IgG3 in IgA immune complexes. In combination with chemoattractants released from mesangial cells, C3a and C5a promote monocyte/macrophage recruitment into the glomerulus.⁸⁵ Local, intrarenal, synthesis of complement proteins may also participate in the inflammatory response in IgAN.^{84,86,87}

Genetic landscape. While familial cases of IgAN have been reported from France, Italy, Australia, the United States, and the Middle East, despite extensive genetic analyses in kindreds, no consistent mutation or pattern of inheritance has been confirmed.^{71,88-92} Interestingly, genetic loci reported from familial cases have failed to be confirmed in the large GWA studies undertaken in sporadic primary IgAN. There are reports of concordance and, importantly, discordance for IgAN between monozygotic twins.⁹³⁻⁹⁷

In the largest meta-GWA study to date, more than 30 different risk loci were identified. This study included 10,146 cases of biopsy-proven IgAN and 28,751 matched controls across 17 international cohorts. The genes identified were *FCRL*, *TNFSF4*, *CFH*, *REL*, *CD28*, *PF4V1/CXCL8*, *IRF4/DUSP22*, *LY86*, *HLA-DR*, *HLA-DQ*, *HLA-DP*, *CCR6*, *DEFA*, *LYN*, *ANXA3*, *TNFSF8/15*, *CARD9*, *REEP3*, *ZMIZ1*, *RELA*, *ETS1*, *IGH*, *ITGAM*, *IRF8*, *TNFSF13*, *TNFSF13B*, *FCAR*, *HORMAD2/LIF*, *FCRL3*, *DUSP22*, *IRF4*, and *PADI4*. The presence of each identified SNP only marginally increases the risk of developing IgAN, with odds ratios in the region of 1.1 to 1.6. Pathway analysis has shown that these risk loci are closely associated with maintenance of mucosal integrity and the intestinal immune network for IgA production.^{63,98-100,103} However, these genetic loci collectively only explain ~11% of the risk of developing IgAN,⁶³ implying that environmental factors drive the majority of the risk.

Microbial encounters and the microbiome. The remaining >80% of risk for developing IgAN must then be derived from the environment, and the most imposing environmental factor is the microbiome. Interactions between the mucosal immune system, the resident microbiome, and environmental pathogens are likely to be major drivers for pathogenic IgA production and the development of IgAN. As already mentioned, IgA and IgG antibodies reactive against dg-IgA1 can be physiologic antimicrobial antibodies generated during appropriate responses to encounters with microbes presenting the Tn antigen.⁷² There is also an increasing body of evidence supporting the role of the microbiome in driving pathogenic IgA production.

Changes in the oropharyngeal and intestinal microbiomes have been reported in IgAN.^{104-107,108-112} *Streptococcus mutans* expressing the collagen binding protein (Cnm) was increased in the saliva of patients with IgAN,¹¹³ and experimentally inducing dental caries in rats with this bacterium induces an IgAN-like glomerular disease.¹¹⁴ In general, α -diversity in microbiota is defined by evaluating

the observed richness (the number of taxa) or evenness (the relative abundances of those taxa) within an average sample from a specific anatomic site. In contrast, β -diversity is quantified by measuring the variability in community composition (the identity of observed taxa) across different samples within the same anatomic site. A European study of the oropharyngeal microbiome found no significant difference in α -diversity, but significantly increased levels of *Haemophilus parainfluenzae* were seen in IgAN patients¹⁰⁴ (an example of β -diversity). Moreover, in a Japanese study, *Neisseria* spp. was increased in the saliva of IgAN patients compared with controls.¹⁰⁶ In a Korean study, IgAN patients had significantly greater abundance of *Ruminococci* spp. in tonsillar swabs compared with controls.¹⁰⁸ There have been far more studies worldwide of the intestinal microbiome in IgAN.¹¹⁵⁻¹²⁷ While many significant differences have been reported in IgAN compared with controls, no consistent pattern of dysbiosis has emerged across populations to date. Most studies are limited by the absence of CKD control groups and sample size. A shotgun metagenomic sequencing study reported that *Ruminococcus gnavus*, *Bacteroides fragilis*, and *Bacteroides plebeius* were increased in IgAN patients compared with healthy subjects. Interestingly, *R. gnavus* is also increased in patients with ankylosing spondylitis,¹²⁸ a secondary cause of IgAN.

Exactly how the intestinal microbiome might influence the generation of pathogenic IgA1 in IgAN is unclear, although two pathways offer intriguing possibilities. The first involves microbial PAMPs directly downregulating O-glycosylation of the IgA1 hinge region in mucosal plasma cells, which would leave it more susceptible to bacterial proteases that can digest IgA1 at its hinge region, facilitating eventual evasion of secretory IgA1 by the microbiome. The second involves microbiome enrichment with mucin-degrading bacteria that produce glycosidases capable of stripping O-glycans from the IgA1 hinge region, a process that could also serve to increase susceptibility of IgA1 to bacterial proteases. It has been demonstrated that patients with IgAN have increased relative abundance in their intestinal microbiota of one such species, *Akkermansia muciniphila*, compared with healthy controls and controls with other causes of CKD. This bacteria can directly deglycosylate the IgA1 hinge region in the intestinal lumen, and the resulting dg-IgA1 has been shown to return back to circulation across the mucosal barrier before depositing in the renal mesangium.⁶⁴ An enrichment of functional pathways involving glycosidases with the potential to deglycosylate the IgA1 hinge region, including β -galactosidases and α -N-acetylgalactosaminidases, has also been reported in a metagenomic study of the intestinal microbiome from IgAN patients.¹²³

CLINICAL PRESENTATION

Patients with IgAN classically present with nephritic syndrome (i.e., hematuria, variable amounts of proteinuria, and hypertension with or without impaired kidney function); however, broad variability around this archetypal clinical phenotype exists. At one end of the spectrum, patients with mild disease, detected early on, present with asymptomatic nonvisible hematuria without proteinuria or kidney dysfunction.¹²⁹ Patients will only be diagnosed at this stage if their urine is tested and if they undergo kidney

biopsy despite an absence of proteinuria or impaired kidney function. At the other end of the spectrum, patients with severe disease, detected much later in the disease course, present with established kidney failure.¹³⁰ Such patients will only be diagnosed if a decision is made to proceed to kidney biopsy despite advanced CKD.

Between these two ends of the spectrum, patients with IgAN can present with varying degrees of hematuria, proteinuria, and kidney dysfunction, depending on disease duration, extent of active glomerular disease, and kidney scarring. Visible (macroscopic) hematuria is a frequent clinical presentation in children and young adults.¹³¹ Up to 50% of patients with IgAN experience one or more episodes of visible hematuria over their disease course, commonly coinciding with an upper respiratory tract infection, known as “synpharyngitic hematuria.”¹³² These episodes can be accompanied by flank pain, thought to be due to stretching of the kidney capsule, or a low-grade fever, thereby mimicking urinary tract infection or urolithiasis. These episodes typically abate with age and are rare in patients older than the age of 40 years.

The distribution of clinical presentations across the spectrum of nephritic phenotypes varies by cohort¹³² due to differences in the following:

- Public health policies (e.g., universal urine testing in schools and in the workplace in Japan and Korea) will detect earlier and milder cases;
- Kidney biopsy accessibility, affordability, and threshold to perform a kidney biopsy. Centers requiring moderate proteinuria before considering a kidney biopsy will fail to diagnose earlier and milder cases;
- Genetic polymorphisms (e.g., Asian ancestry) are associated with a higher frequency of IgAN risk polymorphisms.¹³³

Some less common presentations of IgAN warrant further mention:

- IgAN presenting with nephrotic syndrome: A rare presentation of IgAN, affecting less than 10% of patients in most cohorts, is nephrotic syndrome. This is distinct from the presence of nephrotic-range proteinuria, which is often associated with advanced IgAN and extensive glomerular and tubulointerstitial scarring. The histologic correlate of this clinical presentation is diffuse foot process effacement on electron microscopy (EM), as seen in minimal-change disease (MCD). Mesangial proliferation by light microscopy (LM) may be minimal, but IgA will be identified by immunofluorescence (IF) microscopy, distinguishing this condition from MCD. Whether this phenotype is a true podocytopathic variant of idiopathic IgAN or instead represents two concurrent but independent glomerular diseases remains a matter of debate.¹³⁴
- IgAN with rapid loss of kidney function: Less than 10% of patients with IgAN will develop $\geq 50\%$ loss of eGFR over ≤ 3 months. These patients often present with nephritic syndrome as described earlier, but with more rapid loss of kidney function. It can be the first presentation of IgAN or occur later in the disease course. A kidney biopsy is essential to determine the cause of rapid progression, which in $\sim 50\%$ of cases can be caused by something other than aggravated glomerular inflammation.¹³⁵

When rapid progression is directly due to IgAN there are two principle settings:

- Visible hematuria with renal tubular occlusion by red cell casts and tubulotoxic effects of hemoglobin or other substances released by erythrocytes broken down in the tubular lumen. When kidney biopsy is performed in these cases, the commonest lesion identified is acute tubular necrosis (ATN) often in association with RBC casts. Up to 25% of cases show incomplete kidney function recovery; duration of visible hematuria of more than 7 days has been identified as a risk factor for incomplete recovery.¹³⁶
- An acute inflammatory rapidly progressive glomerulonephritis (RPGN): the kidney biopsy will show the presence of crescents with areas of fibrinoid necrosis and endocapillary hypercellularity. The term “crescentic IgAN” should be avoided because there is no standardized definition, and the presence of crescents in isolation does not inform clinical decision making.

In other cases, rapid progression is not directly due to IgAN or worsening glomerulonephritis, with contributing factors including ATN not due to RBC casts, acute interstitial nephritis, and medication toxicity.¹³⁵

- IgAN presenting with hypertensive emergency: Less than 10% of cases present with severe or “malignant” hypertension as part of a nephritic syndrome. This presentation is associated with worse outcomes.^{137,138} Thrombotic microangiopathy can often be appreciated on renal histology.¹³⁹ Laboratory evidence of an associated microangiopathic hemolytic anemia is rare.

DIAGNOSIS OF IgAN

A kidney biopsy is required to diagnose IgAN. There are no serum or urine biomarkers with sufficient sensitivity and specificity to reliably make the diagnosis less invasively.^{140,141} Histopathologic features of IgAN are described below in Section 6.

If an IgA-dominant glomerular disease is identified, then other causes of IgA-dominant glomerular disease, including secondary causes of IgAN, must be considered before arriving at a diagnosis of idiopathic IgAN.^{1,14,71,142-145}

DIFFERENTIAL DIAGNOSIS

Before performing a kidney biopsy, the differential diagnosis of IgAN is broad and includes any kidney disease presenting with glomerular hematuria and/or proteinuria. A nonexhaustive differential diagnosis includes type IV collagenopathies, such as thin basement membrane disease and Alport syndrome, infection-related GN, membranoproliferative GN (immune-complex, monoclonal immunoglobulin, or complement mediated), lupus nephritis, ANCA vasculitis, or Fabry disease.

Once an IgA-dominant glomerular disease is identified by kidney biopsy, other causes of this histologic pattern of injury with IgA deposition need to be excluded (Table 33.1) and then secondary causes of IgAN need to be sought before arriving at a diagnosis of idiopathic IgAN (Table 33.2). This process involves a thorough clinical assessment (history, physical examination, and laboratory screening) to evaluate for the presence of underlying comorbidities and disease drivers.

The pathway to diagnosing primary IgAN can therefore be viewed as involving three steps:

1. Considering IgAN among the differential diagnosis for a patient presenting with hematuria, proteinuria, and/or reduced kidney function: proceed to kidney biopsy as appropriate.
2. If an IgA-dominant glomerular disease is identified, consider mimics of IgAN such as IgA-dominant infection-related GN, IgA vasculitis, and other causes of glomerular disease with IgA deposition as listed in Table 33.1; if IgAN is confirmed, secondary causes as listed in Table 33.2 should be excluded.
3. If there is an atypical clinical presentation (acute kidney injury, nephrotic syndrome), consider another renal diagnosis with incidental glomerular IgA deposition.

Clinical Relevance

IgA-dominant infection-associated GN is more common in patients >65 years old and in patients with diabetes. The clinical context and relative intensity of C3 deposition compared with IgA on immunofluorescence are key to differentiating this from IgAN.

In secondary IgAN, establishing a causal association with systemic disease is complicated by the fact that mesangial IgA deposition has been identified in up to 20% of otherwise healthy people without clinical features of glomerular disease (hematuria, proteinuria, and/or impaired kidney function). This includes living kidney donors^{146,147} and autopsies.¹⁴⁸ This IgA deposition without overt glomerular disease has been termed “lanthanic IgA.”

For some systemic diseases, a clear plausible underlying pathogenic mechanism for secondary IgAN exists. In the case of chronic liver disease, impaired clearance of IgA-containing immune complexes by the asialoglycoprotein receptor on hepatocytes is thought to lead to glomerular IgA deposition.¹⁴⁹ In the case of celiac disease, ingestion of gliadin can lead to the formation of IgA antigliadin antibodies, with subsequent glomerular deposition of gliadin-antigliadin immune complexes.¹⁵⁰ The central role of IgA in mucosal inflammation explains the association with inflammatory bowel disease. In the case of HIV infection, IgA antibodies reactive against anti-HIV IgG or IgM have been demonstrated in the systemic circulation and in the kidney.¹⁵¹

On the other hand, some glomerular diseases can be seen concurrently with IgAN without any clear pathophysiologic association. For example, while it has been suggested that glomerular lesions caused by diabetic nephropathy favor IgA deposition,^{152,153} the epidemiology support incidental coexistence of these two diseases.¹⁵⁴ Investigation of “Anticoagulant Nephropathy” can lead to a diagnosis of IgAN as systemic anticoagulation can trigger macroscopic hematuria in older patients with previously unrecognized IgAN.²² IgAN has been reported in patients who also have mutated *COL4* genes—it is not clear if collagenopathies predispose to IgAN.^{155,156}

Table 33.1 Differential Diagnosis for Glomerular Disease with IgA Deposition

IgA vasculitis	- Histologically identical to idiopathic IgAN - See Section III for a more detailed discussion of clinical features
Monoclonal gammopathy of renal significance (MGRS)	- Monoclonal IgA-containing deposits less common than monoclonal IgG-containing deposits - Presents histologically as heavy chain deposition disease or proliferative GN with monoclonal immune deposits^{145a}
IgA-dominant infection-related GN	- First described in adults with diabetes and <i>Staphylococcus aureus</i> soft tissue infections: The spectrum of causative organisms and patient phenotypes has since broadened. It is more common >65 years old.^{145b} - Typically presents with acute kidney injury, hematuria, and proteinuria, in the setting of active infection. - Characteristic histologic features include diffuse proliferative GN with neutrophilic infiltration by LM, dominant C3 staining by IF, and subendothelial and (humplike) subepithelial deposits by EM.
Lupus nephritis	Mesangial IgA deposition is seen in about 70% of lupus nephritis cases,^{145c} and IgA-dominant cases have been reported.^{145d}
Membranous nephropathy	Rare cases with subepithelial granular deposition of polyclonal IgA have been reported in Asia.^{145e}
IgA anti-GBM disease	Disease with a circulating IgA isotype anti-GBM autoantibody and linear deposition of IgA is recognized.^{145f}
Incidental codeposition of lanthanic IgA	Given the physiologic role of the renal mesangium in clearance of large molecules from the circulation including polymeric IgA, and the high prevalence of “lanthanic” mesangial IgA deposition, consideration should be given to the possibility of by-stander IgA when deposition is found unexpectedly.

HISTOPATHOLOGIC FINDINGS

LIGHT MICROSCOPY

Glomerular changes seen by light microscopy (LM) in IgAN can range from near normal to a marked mesangioproliferative glomerulonephritis, with or without cellular crescents.¹⁴⁵ Mesangial hypercellularity (M) and mesangial expansion are the most common histopathologic features of IgAN. Mesangial hypercellularity can be focal or diffuse and is defined as >3 cell nuclei within a mesangial segment in a 3-μm paraffin-embedded specimen (Fig. 33.2).¹⁵⁷ Proliferation of mesangial cells themselves, as well as infiltration of leukocytes including both glomerular macrophages¹⁵⁸ and CD3+ T-lymphocytes,¹⁵⁹ account for the increased mesangial cellularity. Noncellular mesangial expansion is accounted for by increased mesangial cells

Table 33.2 Secondary Causes of IgA Nephropathy

Clinical Entity	Clinical and Histologic Features
Liver cirrhosis	- Particularly common in alcoholic cirrhosis, although can occur with other causes of chronic liver disease.^{145g} - In adults, histologic IgA deposition is usually accompanied by few or no clinical signs of glomerular disease.
Inflammatory bowel disease (IBD)	- Crohn's disease and ulcerative colitis accounted for 75% and 25% of IBD-associated IgAN, respectively, in one series.^{145h} - IBD is diagnosed before IgAN in the majority, but IBD may be active or inactive at the time of IgAN diagnosis.
Celiac disease	Glomerular IgA deposition can be identified in up to one-third of patients with celiac disease,¹⁴⁵ⁱ although the disease is clinically manifest in <1%.^{145j} There may also be coexisting dermatitis herpetiformis.^{145k}
HIV infection	Glomerular IgA deposits identified in 8% of patients with HIV in one series: mild urinary abnormalities in all cases.^{145l}
Ankylosing spondylitis Rheumatoid arthritis	- There is a recognized association with IgAN.^{145m,145n} - IgAN is an infrequent extraarticular manifestation.^{145o} The profile of renal complications in RA is evolving with modern treatments.^{145p}
Psoriasis	The risk of IgAN increases with severity of skin disease.^{145q}
Linear IgA bullous dermatosis Sarcoidosis	- Subepidermal bullous skin disease reported can present concurrently with IgAN.^{145r} - Cases reported to present synchronously with diagnosis of sarcoidosis. Granulomatous interstitial nephritis and membranous nephropathy are more common than IgAN, however.^{145s}
Cystic fibrosis	IgAN has been reported in case series of patients with cystic fibrosis.^{145t,145u}
Malignancy	- Mesangial IgA deposition with or without glomerulonephritis has been identified in case reports of patients with hematologic^{145v} and solid organ malignancies.^{145w} - Underlying pathogenic mechanisms are not understood. - Anti-TNFα: Cases of IgAN have been reported,^{145x} notably in the context of anti-TNFα therapy for diseases already associated with IgAN (e.g., IBD). - Checkpoint inhibitors: Interstitial nephritis is the most common renal complication, but associated cases of IgAN have also been reported.^{145y}
Drugs	

production of extracellular matrix proteins, such as periostin and vitronectin.¹⁶⁰

A focal segmental pattern of glomerulosclerosis (S) is also seen in most adult case series.¹⁴⁵ Endocapillary hypercellularity (E) is observed in about one-third of cases¹⁵⁷ due predominantly to the presence of monocytes/macrophages

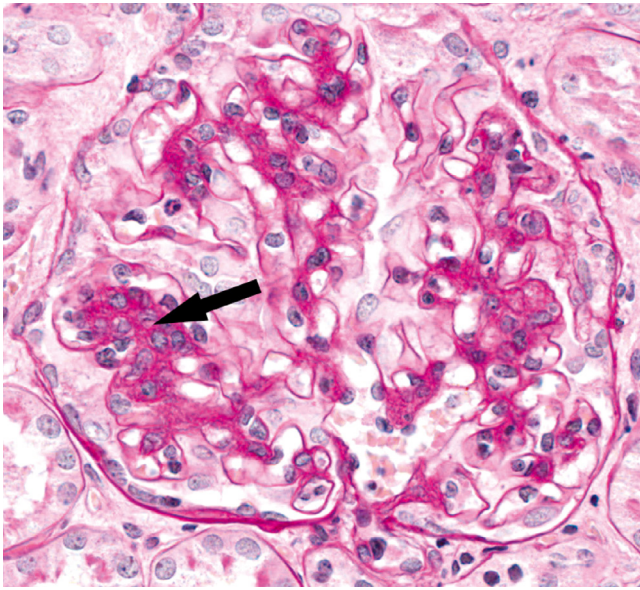


Fig. 33.2 Mesangial hypercellularity in IgAN.

within capillary loops.¹⁵⁷ Glomerular staining for CD68+ (a marker of monocytes/macrophages) correlates with endocapillary hypercellularity but not mesangial hypercellularity.¹⁶¹

Glomerulosclerosis (S) and tubulointerstitial fibrosis (T) are signs of chronic damage. Interstitial fibrosis is associated with mast cells,¹⁶² macrophage, and T-lymphocyte^{163,164} infiltrates.

Cellular, fibrocellular, and fibrous crescents (C) are variably seen, with the lowest frequencies reported in studies of European cohorts (11% of biopsies from the Validation in IgA [VALIGA] cohort¹⁶⁴) and the highest frequency reported in studies of Asian cohorts (~50%–60%).^{165,166} These five features comprise the MEST-C score, which should be reported in all kidney biopsies with a diagnosis of IgAN (Table 33.3).

Thrombotic microangiopathy lesions may also be present and are more common in Asian cohorts. These lesions can appear before the onset of hypertension, consistent with this process contributing to, as well as probably being exacerbated by, hypertension in IgAN.¹⁶⁷

IMMUNOSTAINING

Absolutely necessary for diagnosis of IgAN is dominant or codominant glomerular deposition of IgA. Detection of IgA is usually performed by immunofluorescence (IF) staining; where this is unavailable, immunohistochemistry can be used instead. Deposits of IgA, predominantly in the form of polymeric IgA1,^{44,168} are seen in the glomerular mesangium and can also be seen within the capillary wall in up to a third of cases¹⁵⁷ (Fig. 33.3).

The presence of IgA deposits in capillary walls is associated with the formation of crescents, greater proteinuria, and worse renal outcomes.^{21,143,157,169} A novel monoclonal antibody specific for deglycosylated IgA1 (from a clone named “KM55”) shows positive mesangial staining in primary IgAN and IgA vasculitis but is variably positive in other causes of glomerular disease associated with IgA deposition and therefore is of limited clinical

Table 33.3 MEST-C Score^{166a,166b}

Lesion	Score	Description
Mesangial cell proliferation	M0	Average mesangial hypercellularity score ^a <0.5
	M1	Average mesangial hypercellularity score ^a >0.5
Endocapillary hypercellularity	E0	Absent
	E1	Present
Segmental glomerulosclerosis	S0	Absent
	S1 ^b	Present
Tubulointerstitial fibrosis	T0	≤25%
	T1	26–50%
	T2	>50%
Cellular or fibrocellular crescents	C0	Absent
	C1	Present in at least 1 glomerulus
	C2	Present in >25% glomeruli

^aThe mesangial hypercellularity score is measured as follows: <4 mesangial cells/mesangial area = 0, 4–5 mesangial cells per mesangial area = 1, 6–7 mesangial cells per mesangial area = 2, >8 mesangial cells per mesangial area = 3; the average score of all glomeruli is then calculated.

^bPodocyte hypertrophy/tip lesion characteristics should be reported.

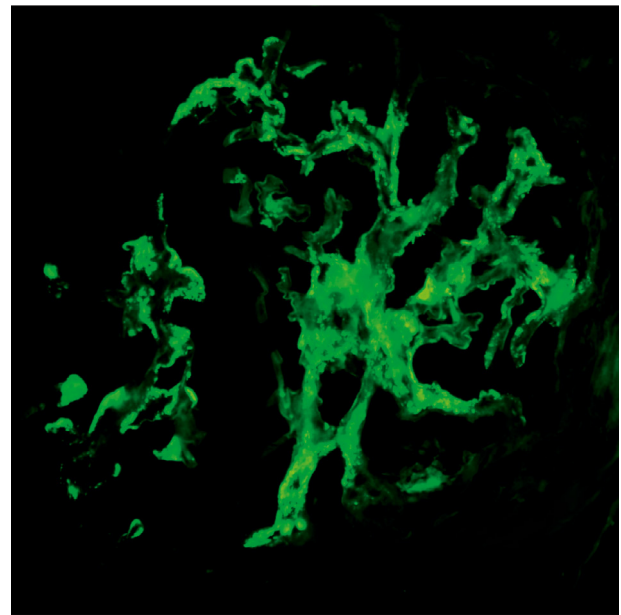


Fig. 33.3 Mesangial IgA deposits detected by immunofluorescent microscopy.

utility.^{170–172} Codeposition of IgG is reported in 15% to 85% of cases and represents IgG autoantibodies targeting dg-IgA1 within immune complexes.^{70,157} IgM deposits are reported in up to 50% of cases.^{169,173,174} The role of IgM in disease pathogenesis is unclear; it has been suggested that IgM deposits simply represent IgM trapping in sclerosed glomeruli. Significantly worse 15-year kidney

survival (60%) has been reported for patients with IgM deposits versus those without (94%), yet these data are likely confounded by the extent of IgM trapping by glomerulosclerosis.^{174,175}

Staining for immunoglobulin light-chains shows a λ -predominance^{169,176} and may even show λ -restriction, despite the absence of a monoclonal gammopathy.

Complement component 3 (C3),^{1,169,173} as well as degradation products of C3, such as C3c and C3d, are codeposited with IgA in almost all cases.^{177,178} Patients with greater intensity of mesangial C3 deposition (2+ or 3+) have significantly worse renal survival compared with those with no C3 deposition.¹⁷⁹ Staining for C1q, representing activation of the classical complement pathway, is almost always negative, distinguishing IgAN from lupus nephritis.¹⁶⁹ C1q positivity has, however, been described in an Asian cohort of IgAN patients, where it is associated with worse outcomes, highlighting the global heterogeneity of the disease.¹⁸⁰

Components specific to the AP are commonly present, with properdin seen in 75% to 100% of cases^{178,181} alongside deposition of factor B.⁸⁴ Deposition of factor H, a negative regulator of the alternative complement pathway, appears to protect against progressive IgAN. By contrast, the extent of deposition of complement factor H related protein 5 (CFHR5), which antagonizes the protective effect of factor H, is associated with a greater likelihood of kidney function decline.¹⁸²

Presence of C4d in the absence of C1q, indicating activation of the lectin complement pathway (LP), is seen in about 40% of cases.¹⁸³ Other components of the lectin complement pathway such as mannose binding lectin (MBL), L-ficolin, and MASP1 and 2 are found in about 25% of cases. A dramatic difference in 20-year renal survival was seen between C4d-positive (28%) and C4d-negative (85%) cases in a retrospective observational study of patients with IgAN in Spain, which remained significant after multivariate analysis.¹⁸⁴ Consistent with this, MBL deposition from LP activation associates with greater proteinuria and kidney failure risk.⁸³ Terminal complement (C5b-C9) deposition is also more frequent in patients with progressive disease.¹⁸³

ELECTRON MICROSCOPY

Electron-dense material is invariably found in the mesangial area,¹⁴⁴ particularly between the contour of the capillary lumen and mesangial space.¹⁸⁵ Deposits are also seen in the paramesangium, which lies between mesangial cells and the glomerular basement membrane (GBM)⁷¹ (Fig. 33.4). Capillary wall deposits are reported in about 40% of cases^{169,186} and can be subepithelial (50%), intramembranous (65%), and/or subendothelial (24%).¹⁸⁶ Capillary wall deposits can sometimes be identified by EM even when IF is negative. The presence of capillary wall deposits is associated with crescent formation.¹⁸⁶

LANTHANIC IgA DEPOSITION

Lanthanic IgA deposition describes positive mesangial staining for IgA without histopathologic evidence of glomerular changes or clinical features of glomerular disease. In autopsy series, mesangial IgA deposition has been identified in approximately 5% of individuals with no known history of kidney disease.^{148,187} For example, a Finnish study reported mesangial IgA deposition among

6.8% ($n = 51$) of 756 autopsies performed on individuals who died from trauma.¹⁸⁸ One of these cases had been diagnosed with IgAN antemortem, and only another nine were suspected to have clinically significant IgAN based on available information. The majority of cases showed isolated IgA deposition without accompanying C3 or IgG.

Further evidence of lanthanic mesangial IgAN comes from biopsies of kidney donors. In a Japanese series, mesangial IgA deposition was present in 82 of 510 (16%) allografts.¹⁴⁷ None of these donors had proteinuria, and only 11 had hematuria >5 RBCs/hpf. In an American cohort, 145/745 (20%) of donor preimplantation biopsies demonstrated mesangial IgA deposition.¹⁴⁶

The high prevalence of glomerular IgA deposits far outweighs the reported prevalence of IgAN, suggesting a high rate of benign mesangial IgA deposition in the general population. The nature of deposited IgA in these subclinical cases has not been explored, but processes associated with IgA deposition in these cases might differ from those that trigger a mesangial cell response. It is important to note that the mesangium plays a physiologic role in the clearance of IgA and other large molecules from the circulation.^{36,75} If lanthanic deposits do have nephritogenic potential, it is estimated that the rate of conversion to clinically identifiable IgAN is 1 in 80.¹⁸⁹

RISK PREDICTION

CLINICAL RISK FACTORS FOR KIDNEY DISEASE PROGRESSION

Many clinical factors have been associated with adverse kidney outcomes. Of these, proteinuria is considered the strongest risk factor for kidney function decline. The risk of progression to kidney failure increases proportionally with incremental increases in time-averaged proteinuria beyond 0.5 g/day. Importantly, reduction in proteinuria through therapeutic intervention can improve kidney outcomes.¹⁹⁰⁻¹⁹² Persistent hematuria during follow-up and increasing amounts of hematuria also predict progression to kidney failure,¹⁹³ while remission of hematuria is associated with a slowing in the rate of GFR loss.¹⁹⁴ Hypertension is another risk factor for kidney function decline.¹⁹⁵ For every mm Hg increase in mean arterial pressure at presentation, eGFR is seen to decline by an additional average of 0.06 mL/min/1.73 m² per year, and for each mmHg increase in mean arterial pressure during follow-up, eGFR is seen to decline on average by an additional 0.24 mL/min/1.73m² per year.¹⁹¹

Beyond proteinuria, hematuria, and hypertension, it is now appreciated that cigarette smoking¹⁹⁶ and obesity¹⁹⁷ are additional clinical predictors of kidney function decline, while Asian ethnicity is a strong demographic risk factor for kidney failure.¹⁸

THE OXFORD CLASSIFICATION

The Oxford Classification was developed to enable consistent reporting of features found in kidney biopsies from IgAN patients that are independently associated with the future risk of kidney function decline. Five histopathologic features are scored, covering lesions associated with acute inflammation (mesangial hypercellularity [M], endocapillary hypercellularity [E], crescents [C]) and

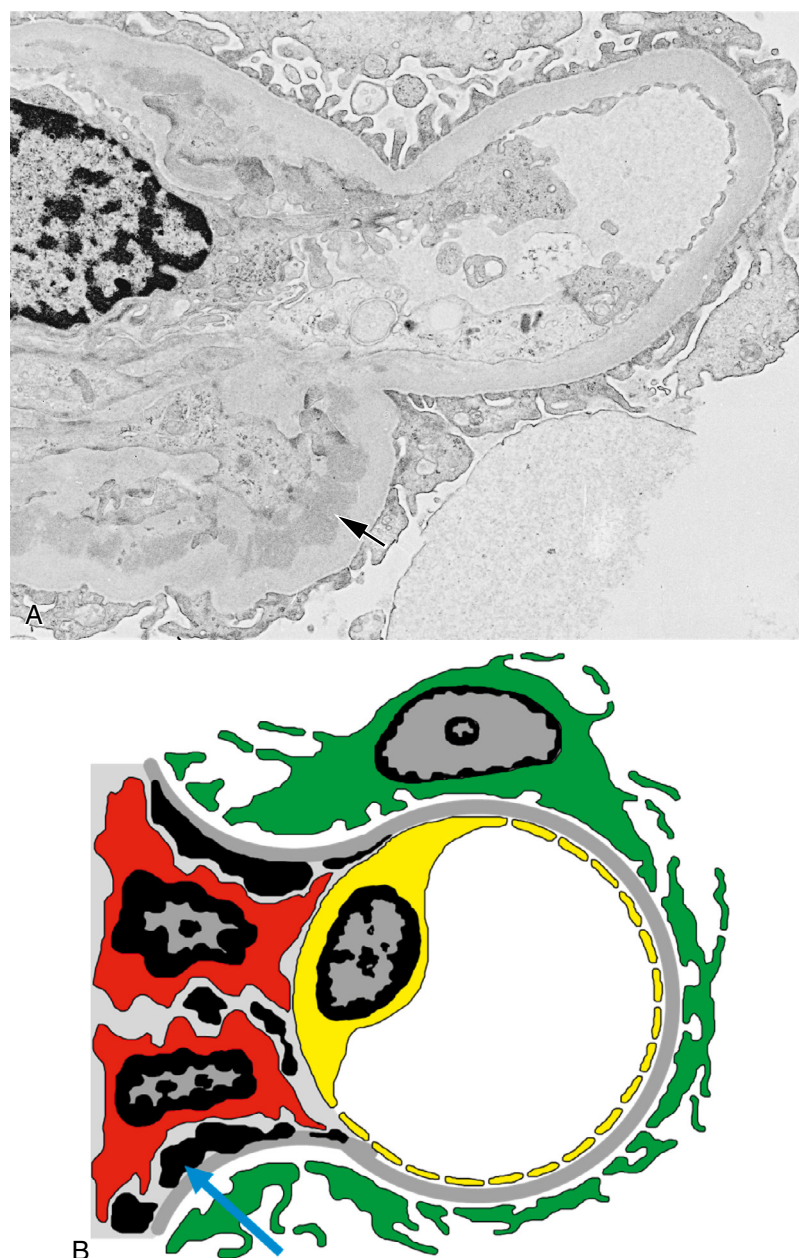


Fig. 33.4 Electron-dense deposits of IgA-containing immune complexes in the paramesangium seen by electron microscopy (A). A cartoon representation of the electron microscopy image is shown in (B).

chronic damage (segmental glomerulosclerosis [S], and tubular atrophy/interstitial fibrosis [T]) (Table 33.4).

Application of this scoring system requires at least eight glomeruli to be present in the sample. In their original report, Cattran and colleagues¹⁹⁸ found that mesangial hypercellularity (M), segmental glomerulosclerosis (S), and tubulointerstitial fibrosis (T) were associated with greater decline in renal function in univariable analyses. Addition of the MEST score to clinical data at biopsy improves prediction of renal outcome and was equivalent to the culmination of 2 years of follow-up clinical data for predicting 5-year renal survival.¹⁹⁹ In a meta-analysis including 3893 patients, M, S, and T-scores independently predict kidney failure.²⁰⁰ The T-score has consistently been shown to be the strongest predictor of poor long-term renal outcome,^{19,200} while the E-score has the weakest prognostic ability, particularly if patients receive immunosuppression. A significant

interaction was found between immunosuppression and endocapillary proliferation (E) with respect to renal outcome in the original Oxford cohort,¹⁹⁸ suggesting that immunosuppression could prevent decline in kidney function when an E lesion was present. The same interaction was seen in the North American validation study²⁰¹ but not in the VALIGA validation study,¹⁶⁴ despite similar use of prebiopsy immunosuppression in these cohorts.

Compared with no crescents, having 1% to 25% glomeruli with crescents increases the risk of kidney failure 1.6-fold while having >25% glomeruli with crescents increases risk by 2.3-fold. On the basis of this, the C score was added to the MEST score in 2017 to form the MEST-C score.²⁰² When IgAN presents as RPGN with >50% crescents on biopsy, it carries a particularly poor prognosis. In a large Chinese cohort, the proportion of such patients who progressed to kidney failure was 47% at 1 year and 70% at 5 years.²⁰⁰

An effect of immunosuppression on the predictive value of the C score has also been shown. This only applies to C1 (<25% crescents), as immunosuppression was not shown to benefit outcomes for C2 (>25% crescents) (see [Table 33.3](#)).^{198,202}

THE INTERNATIONAL IgAN PREDICTION TOOL

An International IgAN prediction tool that incorporates MEST score, GFR, proteinuria, hypertension, ethnicity, and use of immunosuppression or RAASi²⁰³ before kidney biopsy has been validated internationally across multiethnic cohorts and incorporated into international IgAN clinical practice guidelines,¹⁴⁰ as a means to risk-stratify patients at the time of the initial kidney biopsy. Crescents were not found to contribute significantly to the prediction model at time of biopsy; the greatest weight in the model is given to the histologic T-score. This model has a C-statistic of 0.82 for predicting a $\geq 50\%$ decline in eGFR or ESKD and performs marginally better than routine clinical data (eGFR, MAP, and proteinuria at time of biopsy). The variables included in the prediction model are listed in [Table 33.4](#).

POTENTIAL FUTURE BIOMARKERS FOR RISK PREDICTION

A number of biomarkers have been proposed as possible predictive biomarkers in IgAN including serum deglycosylated IgA1^{43,204-206} and the ratio of serum IgA/C3.^{207,208} Increased C3a and C5a levels in the urine and increased expression of their receptors in kidney tissue are associated with the activity and severity of renal injury in IgAN.⁸⁵ Other urinary markers with prognostic potential include the epidermal growth factor (EGF) to monocyte chemotactic peptide-1 (MCP-1) ratio,²⁰⁹ CXCL1,²¹⁰ soluble CD89 (sCD89), and soluble CD163 (sCD163).^{211,212} CD163 is a high-affinity scavenger receptor for the hemoglobin-haptoglobin complex and is a marker for cells of the monocyte/macrophage lineage. However, none of these have been systematically validated in large patient cohorts and are not ready for clinical implementation in the foreseeable future.

PROGNOSIS

QUALITY OF LIFE

IgAN has a negative impact on life participation, particularly for those diagnosed at a young age, with IgAN patients reporting poorer psychological well-being. The diagnosis, as well as its associated medical management, can lead to missed educational or professional opportunities, uncertainty about the future, and feeling vulnerable.²¹³⁻²¹⁵ Patients with glomerular disease including IgAN experience a considerable burden of symptoms even before progression to kidney failure. The most bothersome physical symptoms reported by patients are pain and fatigue.²¹⁶ Edema had the strongest association with poorer quality of life scores.²¹⁷ Data gathered from an IgAN support group on social media found that patients felt communication from their nephrologist was inadequate; there was a perceived lack of information on diet, blood pressure, symptom control, and a lack of clarity around possible treatments.²¹⁸ A poll of patients conducted by the National Kidney Foundation and IgA Nephropathy Foundation of America found that 23%

Table 33.4 International IgAN Prediction Tool Variables^{203a}

Age at time of biopsy
eGFR at time of biopsy
Mean systemic arterial blood pressure at time of biopsy
Proteinuria at time of biopsy (g/day)
Use of RAAS inhibition at time of biopsy
Histologic MEST M-score
Histologic MEST E-score
Histologic MEST S-score
Histologic MEST T-score
Race category: Caucasian, Chinese, Japanese, or Other
Use of immunosuppression preceding biopsy
Interaction effect between proteinuria and T-score
Interaction effect between proteinuria and MAP

Data from Barbour SJ, Coppo R, Zhang H, et al. Evaluating a new international risk-prediction tool in IgA nephropathy. *JAMA Intern Med.* 2019;179:942–952.

of IgAN patients experience anxiety, 18% hopelessness, and 18% depression in relation to coping with their diagnosis; 21% reported that their general daily function is limited by having IgAN.²¹⁹

KIDNEY FAILURE RISK

Small studies on the natural history of IgAN, preceding the widespread introduction of renin-angiotensin-aldosterone system inhibition (RAASi), estimated a 15% to 20% rate of progression to kidney failure 10 years from diagnosis.²¹ Analysis of Western patients from the Oxford Classification study and subsequent VALIGA and North American validation studies ($n = 901$) who were exposed to contemporary treatments, including RAASi (86%) and immunosuppression (36%), reported a 27% risk of kidney disease progression (kidney failure or a 50% decline in eGFR) over 10 years.¹⁹⁹ A single-center follow-up study from Japan reported 16%, 33%, and 50% risks of developing kidney failure at 10, 20, and 30 years, respectively.²⁰ More recently, data from a large U.K. population-based cohort of 2,299 adults and children with IgAN exposed to contemporary treatments reported median kidney survival from diagnosis of only 11.4 years (95% CI 10.5–11.5) with a mean age at kidney failure of 48 years.¹⁹⁰ As noted earlier, entry into the cohorts described is impacted by local urine screening and kidney biopsy practice, with potential for introducing lead-time and immortal-time bias.

While actuarial risk of kidney failure appears broadly similar across geographic regions^{18,21} within a multiethnic cohort from North America, Barbour and colleagues¹⁸ found that people of Pacific-Asian origin had a higher risk of progression to kidney failure or 50% decline in eGFR after multivariable adjustment.

MORTALITY RISK

Patients with IgAN are reported to have a 1.5-fold increased risk of death compared with matched controls over 13 years of follow-up in an Asian cohort ($n = 1364$), and had a 6-year reduction in life expectancy compared with the general population in a Swedish population-based cohort study of

Table 33.5 Treatment Strategies for Patients with IgAN

Clinical Presentation	Treatment	Follow-Up
Isolated nonvisible hematuria ^a	No specific treatment required ^b	Lifelong follow-up and minimum annual BP, eGFR, and UPCR applies to all of the clinical presentations
Episodic visible hematuria ^a	No specific treatment required ^b	
Proteinuria with hematuria	See main text	
Nephrotic syndrome ^c	Treat as minimal-change disease (Chapter 30)	
Rapidly progressive IgAN ^d	Treat with cyclophosphamide and systemic glucocorticoids in accordance with the recommendations for AAV (Chapter 32)	
Malignant hypertension	See Chapter 46	

^aWith normal eGFR.^bA kidney biopsy is often not performed and a presumptive diagnosis of IgAN is often made. It is important in these cases to consider a type IV collagen disorder.^cMesangial IgA deposition with light and electron microscopy features otherwise consistent with a podocytopathy resembling minimal-change disease and proteinuria >3.5 g/day with hypoalbuminemia and edema.^dRapidly progressive IgAN is defined as a 50% decline in eGFR over 3 months, where other causes of rapidly progressive glomerulonephritis (e.g., AAV and anti-GBM disease) and reversible causes (e.g., drug toxicity, common prekidney, and postkidney causes) have been excluded. A kidney biopsy is essential in these cases and will commonly demonstrate mesangial and endocapillary hypercellularity and crescents with areas of focal necrosis. The presence of crescents in a kidney biopsy in the absence of a concomitant change in serum creatinine does not constitute rapidly progressive IgAN; however, these patients require close follow-up to ensure prompt detection of any GFR decline.

AAV, ANCA-associated vasculitis; BP, blood pressure; eGFR, estimated glomerular filtration rate; UPCR, urine protein:creatinine ratio.

more than 20,000 individuals.^{220,221} The burden of excess mortality is attributed primarily to cardiovascular disease and appears greatest in those patients who develop kidney failure.²²²

TRANSPLANTATION AND RISK OF RECURRENCE IN ALLOGRAFT

Compared with patients receiving a kidney transplant for other forms of primary glomerular disease, those with IgAN have excellent overall survival post transplantation. Nevertheless, recurrence of IgAN in the allograft can be as high as 50% after 5 years, and results in significantly shorter graft-survival compared with IgAN patients without recurrence.^{223,224} Maintaining corticosteroids as part of transplant immunosuppression regimens has been reported in some studies to be associated with a decreased risk of allograft IgAN recurrence, as is the use of MMF.²²⁵

TREATMENT

Treatment strategies for patients with IgAN are summarized in Table 33.5. For management of atypical presentations of IgAN, including nephrotic syndrome, rapidly progressive IgAN, and malignant hypertension, please refer to the relevant chapters listed in Table 33.5.

Most adult patients with nonvisible hematuria and proteinuria have already developed chronic kidney disease (CKD), with an eGFR on average between 50 and 60 mL/min at the time of presentation, indicating that they have lost at least 50% of their nephron mass before a nephrologist has the opportunity to intervene.^{142,190} As the average age at presentation is between 30 and 40 years²¹ and typical life expectancy in developed countries is 70 to

80 years,²²⁶ there needs to be an immediate focus by the treating nephrologist on the introduction of therapies to preserve all remaining nephrons if kidney failure is to be avoided.

Treatment should be directed at processes driving continued loss of nephrons. This necessitates two fundamental therapeutic approaches. The **first** is to manage generic intrarenal responses to nephron loss, which include the development of glomerular hyperfiltration, the tubulointerstitial response to persistent proteinuria, and the systemic hypertension.²²⁷ The **second** is to target IgAN-specific pathogenic pathways responsible for production of pathogenic IgA, the formation of IgA immune complexes, glomerular IgA accumulation, and consequent activation of proinflammatory and profibrotic pathways within the kidneys.²²⁸

As most patients already have established CKD at the time of diagnosis, a dual approach is often needed to target both the generic and IgAN-specific drivers of continued nephron loss. Treatments that limit the vicious cycle of generic responses to nephron loss have evolved greatly over the past 5 years. These therapies have the largest safety and efficacy evidence base in IgAN and are recommended in the 2021 Kidney Disease Improving Global Outcome (KDIGO) Clinical Practice Guideline for the Management of IgAN.¹⁴⁰ By contrast, treatments that safely target IgAN-specific pathways have only recently become available, with the first drug for this specific indication being approved by the U.S. Food and Drug Administration in 2021. Over the next 5 years there are likely to be many more drugs approved, offering nephrologists varied new opportunities to specifically target the IgAN pathogenic cascade.²²⁸

MANAGING THE GENERIC RESPONSES TO IgA NEPHROPATHY-INDUCED NEPHRON LOSS

Traditionally this has been referred to as “optimized supportive care,” incorporating lifestyle modification and drugs that minimize glomerular hypertension. Integral to this approach is simultaneously addressing cardiovascular risk factors. Over the past 5 years the range of drugs available to manage the generic response to nephron loss in CKD has expanded significantly.²²⁹

Lifestyle Modification

All patients with IgAN should receive guidance on dietary sodium restriction (<2 g/day, <90 mmol/day), smoking cessation, weight control, and exercise as appropriate.¹⁴⁰ Other than dietary sodium restriction, there is no evidence for benefit of any specific dietary intervention, except gluten avoidance for those with confirmed celiac disease and secondary IgAN.

Addressing lifestyle factors can positively affect systemic and glomerular hypertension and reduce overall cardiovascular risk. However, in patients with IgAN and established CKD, additional drug treatments are almost invariably required to achieve treatment goals.

Management of Systemic Hypertension

An extensive body of evidence shows that uncontrolled systemic hypertension is a major risk factor for progression of CKD and that in proteinuric patients, treatment of systemic hypertension reduces the risk of progression to kidney failure.^{230,231} Data specifically in IgAN, while not extensive, are consistent with these observations.^{191,195} Target systolic blood pressure in most adult IgAN patients should be <120 mm Hg using standardized office blood pressure measurement.¹⁴⁰ When systemic hypertension is associated with proteinuria >0.5 g/day, a renin-angiotensin-aldosterone system inhibitor (RAASi) should be the first agent of choice.^{191,232,233}

Management of Glomerular/Hyperfiltration

Glomerular hypertension with hyperfiltration is most typically manifest by the appearance of increasing albuminuria.²³⁴ Titration of therapy to achieve the maximal reduction in albuminuria is recommended and normalization of total urine protein excretion (<0.3 g/day), which is thought to reflect normalization of glomerular pressure, should be the goal, although this is rarely achieved with currently available treatments.

Renin-angiotensin aldosterone system inhibition. The 2021 KDIGO guidelines recommend that all patients with IgAN and proteinuria >0.5 g/day, irrespective of whether they have systemic hypertension, should be treated with either an angiotensin-converting enzyme inhibitor (ACEi) or angiotensin receptor blocker (ARB) in an effort to normalize glomerular pressure.¹⁴⁰ A meta-analysis of data from 830 patients across 11 randomized controlled clinical trials in IgAN showed that a reduction in proteinuria was associated with a lower risk of doubling serum creatinine, kidney failure, or death. This benefit was consistent across studies and independent of the presence or absence of systemic hypertension.²³⁵

No large clinical trial data are available on the efficacy or safety of dual blockade with an ACEi and ARB specifically in IgAN. A post hoc analysis of the STOP-IgAN trial reported no additional benefit with dual blockade,²³⁶ and the 2021 KDIGO IgAN guidelines advise against this.¹⁴⁰

Sodium-glucose cotransporter-2 inhibition. Multiple large trials have demonstrated a kidney-protective effect of sodium-glucose cotransporter-2 (SGLT2) inhibitors in patients with CKD. This effect is thought to be primarily mediated through natriuresis and reduction in intraglomerular pressure.

The Dapagliflozin and Prevention of Adverse Outcomes in Chronic Kidney Disease (DAPA-CKD) was prematurely terminated at 2.4 years due to efficacy: The dapagliflozin group reached the primary composite outcome (50% decline in eGFR, kidney failure, or death from renal or cardiovascular causes) less frequently than the placebo group.²³⁷ A prespecified subgroup analysis of the 270 patients with IgAN confirmed similar favorable results on eGFR loss and proteinuria in the dapagliflozin arm, with a hazard ratio (HR) of 0.29 (95% CI 0.12–0.73) for the primary outcome.²³⁸ The Study of Heart and Kidney Protection with Empagliflozin (EMPA-KIDNEY trial) was also terminated early at 2.0 years due to efficacy in the SGLT2 inhibitor group.²³⁹ The primary composite outcome of progression of kidney disease or death from cardiovascular cause occurred less frequently in the empagliflozin group compared with placebo (HR 0.72, 95% CI 0.64–0.82), which was equally seen in the 817 patients with IgAN included in the study (HR 0.67, 95% CI 0.46–0.97).²⁴⁰ In a meta-analysis of 13 randomized controlled trials (RCTs) of SGLT2 inhibitors, the risk of kidney disease progression was reduced by 40% (relative risk 0.60, 95% CI 0.46–0.78) in patients with glomerular diseases, and this was similar among disease subcategories of IgAN, focal segmental glomerulosclerosis, and other glomerulonephritides.²⁴¹ Specifically in IgAN, SGLT2 inhibitors reduced the risk of kidney disease progression by 51% (relative risk 0.49, 95% CI 0.32–0.74).

It is, however, important to acknowledge that both DAPA-CKD and EMPA-KIDNEY were CKD trials, not IgAN-focused trials: Baseline eGFR and proteinuria of the included IgAN patients were lower than those reported in IgAN-specific phase 2 and 3 clinical trials. Supportive care including RAASi was not optimized during run-in, and the event rate in the placebo arm in the DAPA-CKD IgAN substudy was higher than expected. Whether SGLT2 inhibition is as effective in those with preserved kidney function before chronic damage is established is not yet clear. Nevertheless, SGLT2 inhibition has emerged as an important addition to supportive care in IgAN.

Endothelin receptor antagonism. Endothelin-1 (ET-1) is a potent regulator of glomerular blood flow through activation of endothelin A (ETA) receptors. Moreover, dysregulation of ET-1 is believed to be a major driver for development of glomerular hypertension and fibrosis in CKD.²⁴² Sparsentan is a dual endothelin angiotensin receptor antagonist (DEARA); it has been shown in the phase 3 PROTECT trial

in IgAN to significantly reduce proteinuria after 36 weeks compared with an optimized dose of the ARB irbesartan. At 2 years, proteinuria remained lower in the sparsentan group and patients in the sparsentan group also had a slower rate of eGFR decline compared with irbesartan (-2.7 mL/min/ 1.73 m² compared with -3.8 mL/min/ 1.73 m²), with no significant differences in adverse events between the groups.²⁴³ The U.S. Food and Drug Administration (FDA) granted full approval to sparsentan for IgAN treatment in September 2024. Preliminary results from the ALIGN phase 3 trial evaluating the efficacy of atrasentan, a sole ETA receptor antagonist, in addition to optimized RASi in IgAN have shown a significant reduction in proteinuria at 9 months compared to RASi alone ([ClinicalTrials.gov Identifier: NCT04573478](https://clinicaltrials.gov/ct2/show/study/NCT04573478)).²⁴⁴

Despite advances over the past 5 years in managing the generic responses to IgAN-induced nephron loss, significant residual risk of IgAN progression persists and outcomes remain poor, as might be expected if IgAN-specific drivers of nephron loss are not simultaneously addressed. Long-term follow-up of patients from the STOP-IgAN trial, where there was an intensive optimization of supportive care, showed that almost half of the patients reached the primary composite endpoint of all-cause mortality, kidney failure, or a 40% decline in eGFR after 7.4 years of follow-up.²⁴⁵ In the SGLT2 inhibitor trials, residual rate of loss of kidney function remained high: In the IgAN cohort of the DAPA-CKD trial, this was 3.6 mL/min/ 1.73 m²/year, while in the PROTECT trial, despite the use of formally optimized supportive care and the addition of an endothelin receptor antagonist, loss of kidney function was 2.7 mL/min/ 1.73 m²/year. The triple combination of RAAS inhibition, SGLT2 inhibition, and endothelin receptor antagonism is the focus of ongoing clinical trials.

MANAGING THE IgA NEPHROPATHY-SPECIFIC DRIVERS FOR NEPHRON LOSS

The key initiators of nephron loss in IgAN are 1. production of pathogenic IgA and formation of IgA immune complexes; 2. inflammatory response to mesangial IgA immune-complex deposits; and 3. profibrotic responses to mesangial IgA immune-complex accumulation. Ultimately, blocking the production of pathogenic IgA would be expected to switch off all downstream pathogenic pathways. However, this is likely to take time and most patients at presentation have already sustained significant nephron loss. Accordingly, an immediate antiinflammatory and antifibrotic approach is desirable alongside starting treatment to stop pathogenic IgA production.

Blocking the Production of Pathogenic IgA

B cell/plasma cell depletion. A successful approach employed in a number of autoimmune diseases, including primary membranous nephropathy and ANCA-associated vasculitis, is to target pathogenic immunoglobulin producing B cells using an anti-CD20 B cell depletion approach. However, a trial of rituximab, an anti-CD20 chimeric monoclonal antibody, demonstrated no benefit in proteinuria reduction, kidney function preservation, or reducing levels of dg-IgA1 in an open-label study in IgAN.²⁴⁶ This may

reflect the inability of rituximab to target tissue-resident mucosal IgA+ plasmablasts, which are CD20⁻ or CD20⁺/CD38⁺ plasma cells.²⁴⁷ Alternative depletion approaches specifically targeting plasma cells are currently being evaluated in IgAN ([Table 33.6](#)). The proteasomal system is crucial for cellular protein turnover, which is essential for maintaining cellular homeostasis. Bortezomib reversibly binds a subunit of the 26S proteasome, inhibiting its function. A small pilot open-label trial of bortezomib, a plasma cell depleting agent, reported that 3 of the 8 patients studied achieved complete remission after 4 doses of bortezomib at 1-year follow-up, suggesting that depletion of plasma cells, rather than CD20+ B cell, could potentially improve outcomes in IgAN, although levels of pathogenic IgA dg-IgA1 were not measured in the study.²⁴⁸ Larger trials are required to confirm efficacy and safety.

An alternative approach, almost exclusively adopted in Japan, is tonsillectomy as a means of depleting a large reservoir of IgA⁺ mucosal B and plasma cells. Evidence for effectiveness of tonsillectomy is restricted to observational studies from Japan^{249,250} and China.²⁵¹ A randomized controlled clinical trial of tonsillectomy in Japan reported an improvement in proteinuria but failed to confirm improved outcome for kidney failure.²⁵² A retrospective observational study of European patients found no difference in renal outcomes between IgAN patients who had undergone tonsillectomy and those who had not.²⁵³

B cell/plasma cell modulation. An association between mucosal inflammation and IgAN is well established.²⁵⁴ The mucosal-associated lymphoid tissue (MALT) of the distal ileum is believed to be a major site of pathogenic IgA production.²⁵⁵ A targeted release formulation of the corticosteroid budesonide (Nefecon) delivers budesonide to the distal ileum, where it has been hypothesized to modulate lymphocytes within the MALT and suppress production of pathogenic IgA.²⁵⁵

The NefIgArd trial demonstrated that 9-month treatment with budesonide (Nefecon) 16 mg/day compared with placebo significantly reduced proteinuria at 9 months and slowed the loss of kidney function over the 2 years of the study.²⁵⁶ Nefecon was generally well tolerated with a more acceptable safety profile compared with systemic glucocorticoid therapy due to limited systemic absorption of budesonide. Nefecon was granted accelerated approval by the FDA in December 2021 and full approval for the treatment of IgAN in December 2023.

A new class of B cell-modulating drugs that block the action of B cell activating factor (BAFF) and A Proliferation Inducing Ligand (APRIL), two critical cytokines involved in IgA class switch recombination and B cell/plasma cell proliferation and survival, are currently being evaluated in a number of phase 2 and 3 clinical trials in IgAN (see [Table 33.6](#)). Sibeprenlimab and zigakibart are monoclonal antibodies that inhibit APRIL, while atacicept, telitacept, and povetacept are fusion proteins containing extracellular portions of the BAFF/APRIL receptor TACI, which inhibit both BAFF and APRIL. Early studies have shown that in contrast to CD20 B cell depletion, APRIL and BAFF/APRIL

Table 33.6 Drugs in Development in 2024 to Target the IgA Nephropathy-Specific Drivers of Nephron Loss

Targeting Pathogenic IgA Synthesis					
Plasma Cell Depletion		Mezagitamab <i>Anti-CD38 Antibody</i> NCT05174221 ^a		Felzartamab <i>Anti-CD38 Antibody</i> NCT05065970	
B cell/ plasma cell modulation	Sibeprenlimab <i>APRIL antagonist</i> NCT05248646	Zigakibart <i>APRIL antagonist</i> NCT05852938	Atacicept <i>BAFF/APRIL antagonist</i> NCT04716231	Telitacicept <i>BAFF/APRIL antagonist</i> NCT05799287	Povetacicept <i>BAFF/APRIL antagonist</i> NCT05732402
Inhibiting Glomerular Inflammation					
Alternative complement pathway inhibition	Iptacopan <i>Factor B</i> NCT03373461	IONIS-FB-LRx <i>Factor B</i> NCT05797610		Vemircopan <i>Factor D</i> NCT05097989	
Terminal complement pathway inhibition	Pegcetacoplan C3 NCT03453619	ARO-C3 C3 NCT05083364		Ravulizumab C5 NCT04564339	Cemdisiran C5 NCT03841448 Avacopan <i>C5a receptor</i> NCT02384317

^aShows [ClinicalTrials.gov](https://clinicaltrials.gov) Identifier.Data from <http://www.clinicaltrials.gov> and www.clinicaltrials.gov. Drugs in Phase 3 clinical trials in 2024 are in **BOLD**.

inhibition significantly reduce levels of dg-IgA1 and proteinuria and stabilize eGFR in the short term.^{257,258}

Hydroxychloroquine (HCQ) is routinely used to manage autoimmune diseases including systemic lupus erythematosus and rheumatoid arthritis. While the mechanism of action of HCQ is poorly defined, there is evidence that HCQ modulates lymphocyte activity through inhibiting antigen presentation and TLR signaling.²⁵⁹ Studies from China have reported short-term benefits with HCQ, but these data have not been replicated in other countries and further studies are required before routine use of HCQ in IgAN can be recommended.^{260,261}

Inhibiting lymphocyte proliferation. As in other immune-mediated glomerulonephritides, a number of antiproliferative immunosuppressants including cyclophosphamide, azathioprine, mycophenolate mofetil (MMF), and calcineurin inhibitors have been evaluated in IgAN. Apart from the patient with rapidly progressive IgAN, there is no evidence to support the use of cyclophosphamide, azathioprine, or calcineurin inhibitors in IgAN.¹⁴⁰ Data regarding MMF are, however, conflicting. Small trials of MMF performed in Caucasian patients have failed to show benefit^{262,263}; however, trials from East Asia have reported reductions in proteinuria and slowing of the rate of loss of kidney function in patients with IgAN of Chinese ethnicity.^{264,265} Why there is an apparent difference in response to MMF between different patient populations remains unclear.

Managing Glomerular Inflammation

A major challenge in managing glomerular inflammation is the inability to accurately measure glomerular inflammation and therefore accurately evaluate response to treatment. Traditional measures, the kidney biopsy, hematuria, and proteinuria have significant limitations. A

biopsy only provides a single point-in-time assessment of the kidney; the presence of hematuria can be highly variable, and quantification of erythrocyturia is not universally standardized. Similarly, proteinuria can be present due to noninflammatory processes such as glomerular hypertension/hyperfiltration and glomerular scarring. New biomarkers are emerging but have yet to be validated in IgAN.¹⁴¹

Systemic glucocorticoids. Consistent with many other inflammatory glomerulonephritides, systemic glucocorticoid therapy has been used in IgAN for many years.²⁶⁶ Glucocorticoid-related toxicity and poor tolerability have, however, been significant concerns for both clinicians and patients.²⁶⁷ This was exemplified in both the STOP-IgAN and Therapeutic Evaluation of Steroids in IgA Nephropathy Global (TESTING) studies, where systemic glucocorticoid toxicity was clearly documented, and in the case of the TESTING study led to premature termination of the study.^{245,268} The potential value of targeting glomerular inflammation is, however, supported by such studies. In the TESTING trial, treatment with methylprednisolone was associated with an early fall in proteinuria and improvement in eGFR, consistent with an immediate antiinflammatory effect. This antiinflammatory effect translated to a slowing in the rate of loss of kidney function over the course of the trial. However, when the systemic glucocorticoid was stopped at 6 months and the antiinflammatory effect was lost, the proteinuria relapsed.²⁶⁹ Lowering systemic corticosteroid dose and providing cotrimoxazole prophylaxis mitigated somewhat against infectious complications while maintaining renal benefits.²⁶⁹

The 2021 KDIGO Clinical Practice Guideline suggests that patients at high risk of progressive loss of kidney function can be considered for a 6-month course of systemic glucocorticoid therapy, but only after a thorough toxicity

risk assessment, and it is acknowledged that systemic glucocorticoids should be given with extreme caution or avoided entirely in patients at high risk of adverse effects.¹⁴⁰

Complement inhibition. The poor tolerability and significant toxicity associated with systemic glucocorticoid use, as well as the inability to give repeated treatment cycles due to cumulative toxicity, have led to the development of alternative antiinflammatory approaches in IgAN. Activation of the complement system by IgA immune complexes is thought to be a major driver of glomerular inflammation in IgAN. IgA-induced activation of the complement system occurs predominantly through the alternative and LP, and drugs inhibiting the alternative and terminal complement pathway are being evaluated as novel antiinflammatory therapies in IgAN (see Table 33.6). The most advanced of these is iptacopan, an oral small molecule inhibitor of the alternative pathway complement protein, factor B. In a phase 2 study of 66 patients, iptacopan reduced proteinuria compared with placebo, with no serious treatment-related adverse events reported.²⁷⁰ A phase III study has reported its 9-month proteinuria data, confirming that iptacopan in addition to optimized supportive care significantly reduced proteinuria compared with supportive care alone. Iptacopan is now approved by the FDA for the treatment of IgAN at significant risk of progression. Cemdisiran, an RNA interference therapeutic that blocks production of C5, has also been shown to reduce proteinuria, as has ravulizumab, a C5-targeting monoclonal antibody.²⁷¹ A phase III trial of ravulizumab in IgAN is ongoing.

MANAGING GLOMERULAR FIBROSIS

While there has been a greater understanding of the pathways driving glomerular and tubulointerstitial fibrosis in kidney disease, there has been little progress in translating this improved understanding into novel antifibrotic drugs, and in 2024 there are no antifibrotic drugs on the horizon for use in IgAN.²⁷²

PRESENT AND FUTURE PARADIGMS FOR THE TREATMENT OF IgA NEPHROPATHY

The 2021 KDIGO treatment guidelines recommended that all patients with IgAN are commenced on goal-directed supportive care before disease-modifying therapies are considered. After a period of 3 months of stable optimized supportive care, the risk of progressive kidney function loss should be assessed using residual protein excretion as the biomarker to determine whether additional intervention is required. If urine protein excretion remains >1 g/day, the patient should be offered the opportunity to enroll in a clinical trial. If a trial is not available or the patient declines/is not eligible then, after a thorough toxicity risk assessment, appropriate patients may be offered a single 6-month course of systemic glucocorticoids (Fig. 33.5).

With the approval of new, safer, and better-tolerated therapies, as well as a greater appreciation of the lifetime risk of kidney failure, even with residual proteinuria levels <1 g/day on optimized supportive care, the current treatment paradigm is coming under increasing scrutiny. Future treatment strategies are likely to focus on delivery of a multitargeted approach addressing all factors driving nephron loss in IgAN (Fig. 33.6). There is also likely to be a much lower threshold for

commencing disease-modifying interventions that address the IgAN-specific drivers of nephron loss and these are likely to be commenced at the same time as supportive care interventions, rather than being delayed, as is currently suggested in clinical guidelines. Clinical practice is likely to evolve rapidly over the coming 3 to 5 years as new drugs are approved. Critical to a reasoned and personalized approach to the treatment of IgAN in the coming years will be the introduction into clinical practice of validated biomarkers to allow us to select the right treatment(s), for the right patient, at the right time in their disease course. This will be particularly important when novel therapies, which are likely to be expensive and have distinct safety profiles, are combined to maximize nephron preservation. New draft KDIGO guidelines for the management of IgAN/IgAV are undergoing public review and are expected to be finalized by early 2025.

IgAN IN CHILDREN

INTRODUCTION

As in adults, IgAN in childhood often manifests initially as nonvisible (microscopic) hematuria and can go undiagnosed for many years. Childhood IgAN (cIgAN) was long considered a benign disease^{273,274} subject to spontaneous remission and, occasionally, relapses later in adulthood.²⁷⁵⁻²⁷⁷ However, it has been shown that 30% to 40% of individuals with cIgAN will develop kidney failure, needing kidney replacement therapy within 20 to 30 years of diagnosis.²⁷⁵⁻²⁷⁷ Conversely, in a cohort of 281 children identified through school urine screening programs post-1990 and receiving optimized supportive care, a remarkable 15-year kidney survival rate of 98.8% has been documented, although no account was made for lead-time bias.²⁷⁸ This apparent disparity in outcomes may be influenced by various factors, with ethnicity being notably influential.¹⁵

Despite shared clinical features with adult IgAN, the kidney biopsy in children with IgAN typically shows more proliferative glomerular lesions and fewer sclerotic lesions, although it must be acknowledged that the threshold to perform a kidney biopsy in children is very different from that used in adults.²⁷⁹ Children do, however, present more often with acute nephritic/nephrotic syndrome and acute kidney injury than adults.²⁸⁰ The extent of proteinuria and kidney dysfunction both at presentation and during follow-up vary.²⁸¹ Overall, while some children will achieve complete and permanent remission, most children will experience a slow decline in kidney function continuing into adulthood.

EPIDEMIOLOGY

The reported incidence and prevalence of cIgAN, like adult IgAN, is heavily influenced by the accessibility, affordability, and threshold to perform a kidney biopsy. Accepting these confounders, cIgAN is one of the most common primary glomerulonephritides in children worldwide, with an estimated annual incidence of 2 to 10 cases per 100,000 children.^{8,282} cIgAN is detected in 32% of pediatric cases of glomerular disease in Korea²⁸³ and 40% in Japan according to their respective national

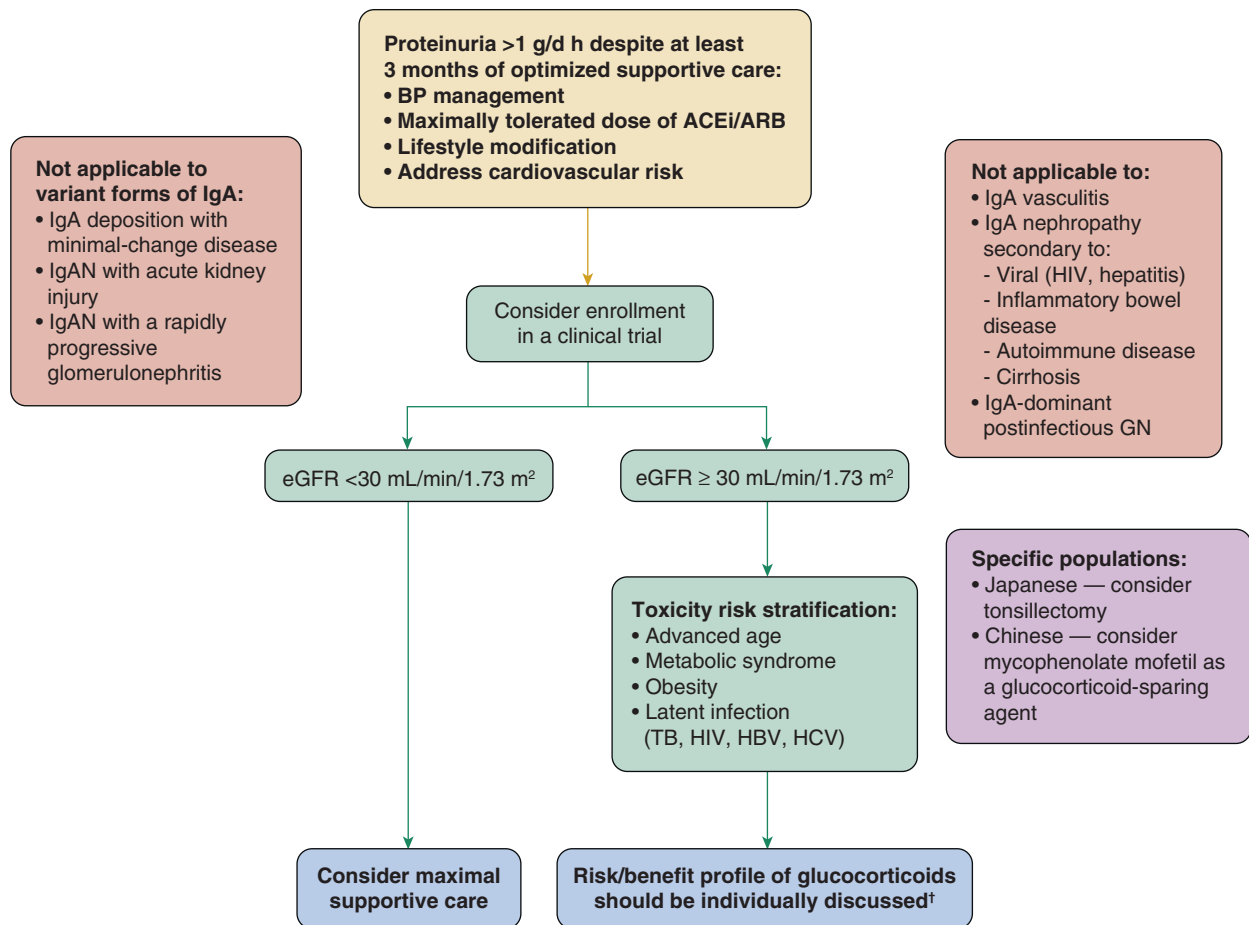


Fig. 33.5 Management of patients with IgAN who remain at high risk for progression after maximal supportive care according to the KDIGO 2021 Clinical Practice Guideline for the Management of Glomerular Diseases. ACEi, Angiotensin-converting enzyme inhibitor; ARB, angiotensin II receptor blocker; BP, blood pressure; eGFR, estimated glomerular filtration rate; GN, glomerulonephritis; HBV, hepatitis B virus; HCV, hepatitis C virus; HIV, human immunodeficiency virus; IgAN, immunoglobulin A nephropathy; TB, tuberculosis. (From KDIGO 2021.¹⁴⁰)

biopsy registries.^{283,284} By contrast, 10% to 26% of pediatric biopsies in case series in Europe,²⁸⁵ 20% in China,²⁸⁶ 14.5% in India and South America, and only 2.8% in Africa have a diagnosis of IgAN.¹⁶ These variations can partly be explained by differences in national screening policies and lower thresholds for kidney biopsy in parts of Asia. In Japan, all schoolchildren between the ages of 6 and 18 undergo annual urinalysis screening. In Japan and South Korea investigation of isolated asymptomatic hematuria includes kidney biopsy, unlike most of the rest of the world, where proteinuria or impaired kidney function is also typically required to justify a kidney biopsy. Identification of asymptomatic urinary abnormalities and performing of kidney biopsies in children is heavily influenced by affordability and accessibility to health care, particularly in low-income countries with limited health care resources.²⁸⁴ Childhood IgAN is more commonly observed moving from west to east, as described in adults.

As might be expected, in countries with active screening programs clinical suspicion of cIgAN is most commonly based on an abnormal urinalysis. A study in Japan, involving 374,846 children, identified 37 cases of IgAN with an average age at diagnosis of 10.7 years. Of these cases of cIgAN, 75.7% were biopsied due to abnormal urinalysis findings during school

screening, while the rest presented with visible hematuria; none presented with nephrotic syndrome or acute kidney injury.²⁸² In a Japanese cohort of 258 children with IgAN, 62% had nonvisible hematuria with or without proteinuria, 26% had episode(s) of visible hematuria, and only 12% presented with acute nephritic or nephrotic syndromes.²⁸⁷ Conversely, in countries without urinary screening, cIgAN is more often identified after an episode of visible hematuria, which is usually associated with an infectious episode, typically an upper respiratory tract infection.^{288,289} A large series from Spain ($n = 939$) reported IgAN in 11.6% of pediatric kidney biopsies, with visible hematuria as the presenting symptom in 50.5% of these cases.²⁹⁰

PATHOPHYSIOLOGY

The pathophysiology of IgAN in childhood closely resembles that described for adults.²⁹¹⁻²⁹⁴ Similar changes in the serum levels of dg-IgA²⁹⁵ and IgA-reactive IgG antibodies²⁹¹ have been reported in children. However, there are reports of differences in the composition of circulating immune complexes in children: cIgAN displays unique features including the presence of soluble sCD89, that correlates strongly with glomerular inflammation.²⁹¹

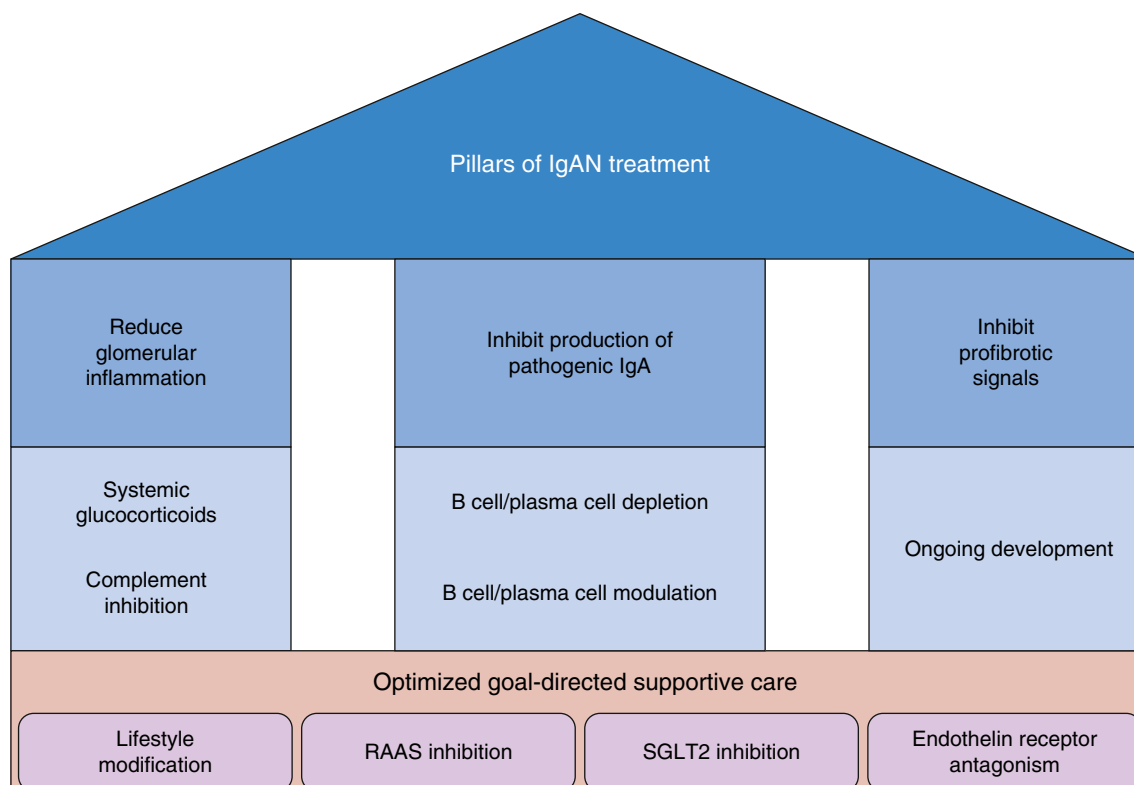


Fig. 33.6 Pillars of IgAN treatment. RAAS, Renin-angiotensin-aldosterone system; SGLT2, sodium-glucose cotransporter-2.

As in adults, the mucosal immune system (MALT) is believed to play a significant pathogenic role in children,²⁹⁶ exemplified by the association between mucosal infections, particularly acute tonsillitis, and episodes of visible hematuria.²⁸⁹ Children are more susceptible to recurrent episodes of acute tonsillitis, and in children predisposed to developing IgAN, these ear, nose, and throat infections often serve as the trigger for visible hematuria.^{144,282} Like adults, children with IgAN exhibit elevated levels of IgA antibodies directed against alimentary antigens, which may be coupled with increased intestinal permeability.²⁹⁷ Peyer patches—the principal induction site for B cell induction and IgA class switch recombination in the gut-associated lymphoid tissue (GALT)—peak during adolescence, when they expand to double the size found later in adulthood.²⁹⁸ In parallel, other lymphoid tissue, including the thymus and tonsils, experience peak growth by late childhood (12–13 years of age), followed by involution during adulthood.²⁹⁹

CLINICAL PRESENTATION

Any form of glomerular syndrome is possible in cIgAN, but the presentation is dominated by hematuria with different amounts of proteinuria:

Isolated, asymptomatic nonvisible hematuria is often identified by chance during a routine examination (in countries without dedicated screening programs). Children will usually not develop progressive CKD during childhood, although the extent of nonvisible hematuria may vary over time.^{300,301} However, glomerular proliferative lesions have been reported in some case series of cIgAN with isolated nonvisible hematuria,³⁰² suggesting a potential risk with this presentation for glomerulosclerosis and CKD development in adulthood.

Recurrent visible hematuria is a more frequent presentation in children versus adults, typically occurring within 48 hours of an upper respiratory tract infection (synpharyngitic hematuria) or episode of gastroenteritis. The episode of visible hematuria generally resolves within a few days, but some children will have prolonged hematuria, which can be associated with irreversible kidney damage when it is associated with significant proteinuria.³⁰³ While recurrent gross hematuria without proteinuria in cIgAN has been associated with a favorable prognosis, recent analyses have revealed the presence of acute and chronic inflammatory changes on kidney biopsy of such patients.³⁰² These episodes of recurrent visible hematuria can be accompanied by obstructive acute kidney injury due to the formation of red blood cell casts when hematuria is abundant, as described in a Chinese cohort where 9.7% of cIgAN had acute kidney injury due to occlusive red blood cell casts, with associated endocapillary hypercellularity and crescents.³⁰⁴

NEPHROTIC SYNDROME ASSOCIATED WITH HEMATURIA

This presentation is more common in children than adults (e.g., 7%–10% in a French cohort²⁸⁹ and 7% in a Japanese cohort of 426 Japanese children and adolescents)³⁰⁵ and is associated with a high risk of progression to kidney failure³⁰³ if not managed appropriately. Kidney biopsy commonly shows severe proliferative lesions with mesangial, endocapillary, and extracapillary hypercellularity.^{289,305} Such cases have a five-fold higher risk of progressive kidney disease compared to cases of cIgAN without nephrotic syndrome at the time of diagnosis.³⁰⁵

A separate group of children will have kidney biopsy features more closely resembling minimal-change disease

(MCD) with mesangial IgA deposits, isolated mesangial proliferation without endocapillary proliferation, and diffuse podocyte foot process effacement on EM.³⁰⁶ Whether this represents a podocytopathic variant of IgAN or two glomerular diseases occurring simultaneously is unclear.³⁰⁶ This presentation is rare, occurring in 2.8% to 8% of cIgAN cases.³⁰⁶⁻³⁰⁸ Many of these cases behave identically to steroid-sensitive nephrotic syndrome, achieving prompt complete remission with glucocorticoid therapy and similar outcomes to children with typical steroid-responsive minimal-change nephrotic syndrome.³⁰⁹ In accordance with the KDIGO guidelines, management protocols for this condition align with those recommended for isolated MCD.³¹⁰

RAPIDLY PROGRESSIVE IgAN

The definition in children varies in the literature but commonly encompasses a rapid decline in kidney function (between a few days and a few weeks) with hematuria and proteinuria and more than 50% crescents on biopsy. This represents <5% of cases (more frequent in adolescents and adults), with kidney failure developing over weeks to a few months if untreated. In children, the histopathologic features closely resemble those of vasculitis with fibrinoid necrosis, arteriolar damage, and cellular and fibrous crescents.³¹¹

DIAGNOSIS

During the initial episode, it is crucial to measure levels of complement C3 and C4 as part of screening for acute postinfectious glomerulonephritis, which is the main differential diagnosis. However, the definitive diagnosis of cIgAN relies on kidney biopsy findings, specifically the presence of dominant or codominant IgA deposits in the mesangium.

HISTOLOGY

Although principally developed for adults, the Oxford classification system using the MEST-C score is commonly applied to cIgAN. However, the initial Oxford study included only 56 children and subsequent validation studies in pediatric populations have found discordant results.³¹²⁻³¹⁴ This might be explained by the fact that children are treated with immunosuppression more commonly, and the much smaller pediatric cohorts studied.³¹⁵ Moreover, histology between pediatric and adult IgAN is different. One clear difference between adults and children is the significance of the S lesion. Most commonly the S lesion develops as a consequence of nephron loss, arterial hypertension, and obesity, which are characteristics frequently encountered in adults and rarely seen in children. Recognition of the podocytopathic subcomponent of the S lesion in IgAN is particularly relevant to children.^{202,289} Importantly, the Oxford classification holds significant relevance for pediatric treatment decisions, as highlighted in the 2021 KDIGO guidelines.

PROGNOSIS

cIgAN is associated with substantial morbidity. In Japan, among children treated between 1976 and 1989 before widespread clinical use of immunosuppressive treatment, reflecting the natural history of the disease, 31% of the cohort progressed to ESKD over a 20-year follow-up period.

In contrast, among a cohort from the 1990 to 2004 period, when RASi and steroids were more commonly prescribed, only 1.2% progressed to ESKD over a 15-year follow-up period.²⁷⁸ In a more recent Japanese cIgAN cohort of 241 children, 5% of individuals developed an eGFR <30 mL/min/1.73 m² within 5 years of disease onset, 6% within 10 years, and 11% within 15 years.³¹⁶ A study of 103 children with IgAN in Kentucky and Tennessee in the United States reported kidney survival rate of 87% at 10 years and 70% at 20 years post diagnosis, compared with a 10-year kidney survival rate of 78% in adults in the same region.^{275,317} A Swedish study of 34 children with IgAN reported that after 8- to 14-year follow-up, 9% had a reduced eGFR while 53% had no evidence of kidney disease (normal urinalysis, serum creatinine, and blood pressure) at last follow-up.³¹⁸ cIgAN diagnosed before age 16 generally exhibits a more benign course than in those diagnosed at ages 16 to 17.²⁷⁷ These findings might suggest a potentially milder manifestation of IgAN in younger children, but these data are likely to be confounded by differences in clinical thresholds for kidney biopsy and treatment paradigms (including medication adherence in adolescents), alongside the impact of an early versus delayed diagnosis.

A number of studies have shown that at presentation, children exhibit more proliferative lesions with fewer sclerotic lesions compared with adults with IgAN, despite similar levels of proteinuria.^{279,319} In children, the proteinuria is likely driven by acute inflammatory glomerular lesions (M1, E1, and C1) while in adults, chronic sclerotic/fibrotic lesions (S1 and T1) and nephron reduction likely play a more dominant role.^{280,319,320} On this basis it has been proposed that adult IgAN often represents a delayed and more advanced presentation of IgAN that originated in childhood but was not detected.

RISK PREDICTION

Assessment of progression risk is necessary to guide treatment decisions in cIgAN, where more potential life-years warrant timely and precise intervention. The International IgA nephropathy Network Pediatric Prediction Tool, available at <https://qxmd.com>, has undergone validation using a multiethnic international cohort of 1060 children.³²¹ The factors included in the pediatric prediction tool are the same as those used in the adult prediction tool (see Table 33.4). However, the precise relationship between these factors and long-term outcomes are less clear-cut for children than adults.³⁰³ Notably, the eGFR trajectory in children contrasts with that of adults, characterized by a highly nonlinear pattern with an initial increase until 18 years of age, followed by a linear decline mirroring that of adults. The pediatric prediction tool can be used to predict the risk of a 30% decline in eGFR or ESRD after biopsy.

As in adults, the extent of proteinuria has a strong association with the rate of loss of kidney function in children. Among 174 children in the European VALIGA cohort, time-averaged proteinuria was a significant prognostic factor in children with proteinuria >0.4 g/day/1.73 m² increasing the risk of IgAN progression.³¹⁵ Achieving an early remission of proteinuria has been linked to a lower risk of doubling serum creatinine, kidney failure, or death.³²²

The Oxford classification is an important component of the pediatric prediction tool; however, there is debate on the extrapolation of kidney biopsy features in adults to children. In a French pediatric cohort, podocytopathic features on kidney biopsy were the sole pathologic parameter predicting kidney function decline, irrespective of clinical data at the time of biopsy and subsequent use of immunosuppression.²⁸⁰ Future subclassification of S lesions (with or without podocytopathy) and inclusion in the pediatric prediction tool could improve risk precision and help inform treatment.^{289,323}

An area of continued debate is the prognostic value of nonvisible hematuria. Persistence of hematuria may indicate ongoing disease activity,³²⁴ and resolution of hematuria has been associated with improved kidney survival in cIgAN.^{194,303}

As in adults, there is a significant need for validated noninvasive biomarkers to assess disease activity and response to treatment in cIgAN. DgIgA1 and IgGdg IgA1 complexes do not correlate with disease activity in cIgAN.²⁹¹⁻²⁹⁴ By contrast, levels of soluble CD89, either free or complexed with IgA1,²⁹¹ and the ratio of IgA/C3 have been associated with a poor prognosis in children.³²⁵⁻³²⁷ Pathologic heterozygous variants in *COL4A3* predispose to a more severe cIgAN presentation.¹⁵⁵ Thin basement membrane disease has been reported in 15% of cIgAN cases,³²⁸ and the *COL4A4* and *COL4A3* genes are situated at 2q36a genetic a genetic locus found to cosegregate with familial cases of IgAN.³²⁹

TREATMENT

Since children have more potential life-years, there is a need for early targeted treatment of IgAN diagnosed in childhood to preserve kidney function for longer. As discussed in the adult section, treatment should immediately address the key drivers of continued nephron loss. Currently, data are insufficient for KDIGO to give specific recommendations regarding the management of children with IgAN, so guidelines are mostly based on those for adults.¹⁴⁰ It is important to acknowledge that children have largely been excluded from the rapid expansion of clinical trials for new therapies in IgAN. There currently is, however, a significant effort by pediatric and adult nephrologists to make these new therapies available to children as quickly as possible.³³⁰

MANAGING THE GENERIC RESPONSES TO IgA NEPHROPATHY-INDUCED NEPHRON LOSS

Optimized Supportive Care

The KDIGO 2021 guideline states that cIgAN with proteinuria >200 mg/day or PCR >200 mg/g should receive renin-angiotensin system blockers (RASB), with dietary sodium restriction, lifestyle modification, and blood pressure control.¹⁴⁰ A randomized controlled trial by Coppo and colleagues³³¹ demonstrated the benefit of angiotensin-converting enzyme inhibition (ACE-I) treatment in reducing the progression of kidney damage in young patients with IgAN. Sixty-six children with cIgAN were randomized to either ACE-I or a placebo and monitored for a median duration of 38 months. A consistent, partial remission of proteinuria (<0.5 g/day per 1.73 m²) in 13 (40.6%) of 32

children in the ACE-I group, in contrast to three (8.8%) out of 34 in the placebo group, was observed. Additionally, total remission of proteinuria (<160 mg/day per 1.73 m² for >6 months) occurred in 12.5% of children treated with ACE-I, while none in the placebo group achieved this status. A multivariate Cox analysis demonstrated that ACE-I treatment independently determined outcome. In a prospective, single-arm study, Nakanishi and colleagues³³² showed that over a 2-year treatment period, lisinopril treatment significantly reduced proteinuria levels compared with baseline in 40 children with IgAN and focal mesangial proliferation. Dizziness was observed in 12.5% of children, but these symptoms resolved with a dose reduction.³³² As with adults, there appears to be no advantage of dual RASB over monotherapy in cIgAN. In an RCT comparing lisinopril monotherapy with a combination of lisinopril and losartan, there was a similar frequency of complete remission of proteinuria (89.0% and 89.3%) over 2 years in children with IgAN and focal mesangial proliferation.³³³ The combination group did, however, experience a higher incidence of dizziness. There are limited data on long-term outcomes in children. Higa and colleagues³³⁴ evaluated 32 pediatric IgAN patients with proteinuria of 0.5 g/day/1.73 m² or less, treated with RASB, and reported 100% kidney survival at 15 years.

MANAGING THE IgA NEPHROPATHY-SPECIFIC DRIVERS FOR NEPHRON LOSS

Blocking the Production of Pathogenic IgA

B cell/plasma cell depletion. While there have been reports of benefit of anti-CD20 B cell depletion with rituximab in children with IgA vasculitis,³³⁵ data in cIgAN are limited to a case report of an 8-year-old Caucasian boy with rapidly progressive IgAN who showed significant improvement in proteinuria, renal function, and histologic characteristics following rituximab treatment combined with methylprednisolone, immunoadsorption, cyclophosphamide, and eculizumab.³³⁶ Current trials of anti-CD38 plasma cell depletion in IgAN do not include children (see Table 33.6, adult section).

B cell/plasma cell modulation. The targeted release formulation of budesonide (Nefecon), which is approved for the treatment of adults with IgAN in the United States and Europe, has not yet been studied in children. Venettacci and colleagues,^{337,338} however, reported a successful outcome with enteric budesonide in a 12-year-old boy with IgAN, demonstrating potential utility of Nefecon in cIgAN. The new class of B cell–modulating drugs that block the action of BAFF and APRIL has not yet been evaluated in children.

Inhibiting lymphocyte proliferation.

Mycophenolate mofetil. Mycophenolate mofetil has been evaluated in a placebo-controlled RCT involving children, young adults, and adults in addition to supportive care. The study was prematurely terminated due to a lack of observed benefit.²⁶²

Cyclophosphamide. As in adults, cyclophosphamide is a treatment option in those children with rapidly progressive IgAN.³¹⁰ Jiang and colleagues³³⁹ showed in 11 children with rapidly progressive IgAN that use of cyclophosphamide, in

addition to prednisone resulted in stabilization of kidney function and significantly reduced hematuria, proteinuria, mesangial IgA deposition, and a kidney pathologic activity index. Over the 18 to 60 months post treatment, 6 patients achieved complete remission, while the remaining 5 exhibited a “markedly effective” response. Repeat kidney biopsies revealed diminished mesangial proliferation, significantly reduced crescent formation, decreased segmental sclerosis, and a notable reduction in mesangial IgA deposition. Niaudet and colleagues³⁴⁰ also reported a benefit of cyclophosphamide for rapidly progressive cIgAN. There are no prospective randomized studies examining cyclophosphamide in cIgAN.

Plasma exchange. As an adjunctive strategy, plasma exchange has been employed to rapidly remove pathogenic IgA from the circulation. Data on efficacy are limited to single-center retrospective studies in small numbers of children with rapidly progressive IgAN.³⁴¹ Immunoabsorption has also been employed in single cases of rapidly progressive IgAN.³³⁶ Whether plasma exchange or immunoabsorption offers any benefit in cIgAN is at present not known.

MANAGING GLOMERULAR INFLAMMATION

Systemic Glucocorticoids

Expert consensus in the KIDGO guidelines is that most pediatric nephrologists would treat IgAN with proteinuria >1 g/day or PCR >1 g/g (100 mg/mmol) and/or mesangial hypercellularity, with systemic glucocorticoids in addition to RASB from the time of diagnosis.³¹⁰ Glucocorticoid therapy is reported to be beneficial in retrospective cohort studies in children with glomerular inflammation and proliferative lesions. In a retrospective French cohort, despite differences in baseline proteinuria and time from onset to treatment, those children treated with glucocorticoids and RASB had an improvement in eGFR and a reduction in proteinuria over a short follow-up period of 6 months compared with those treated with RASB alone.²⁸⁹ Similar data have been reported from retrospective studies in Japan.³⁴²

In a Japanese study involving 78 children with IgAN and mesangial hypercellularity, children were randomized to receive prednisolone, azathioprine, heparin-warfarin, dipyridamole or heparin-warfarin, and dipyridamole only—children in the group treated with additional azathioprine and prednisolone had less proteinuria, lower serum IgA levels, and less progression of glomerulosclerosis over the 2-year study period.³⁴³ Ten-year follow-up of this cohort revealed that children who received azathioprine and prednisolone had a lower incidence of reaching an eGFR <60 mL/min/1.73 m². Two out of 40 patients (5%) in the azathioprine- and prednisolone-treated group and 5 of 34 patients (14.7%) in the comparator group developed kidney failure. The 10-year kidney survival probability for each group was 97.1% and 84.8%, respectively.³⁴³

It is important to acknowledge that there have been no RCTs of glucocorticoid monotherapy in cIgAN, and current evidence for efficacy is limited to small retrospective cohorts that are likely heavily confounded. It also needs to be recognized that glucocorticoids are associated with a myriad of toxic effects, which are particularly pronounced in children. The most frequent adverse effects in children

include weight gain, growth retardation, and cushingoid features, while susceptibility to infections remains a serious concern.³⁴⁴ These adverse effects emphasize the need for new alternatives to glucocorticoids.

Complement inhibition. There is a great deal of interest in complement inhibition as a novel antiinflammatory strategy in IgAN, and this is likely to be equally relevant in cIgAN. There have been three case reports in children describing a beneficial effect of C5 inhibition with eculizumab in progressive cIgAN.^{336,345,346} At present, children are excluded from clinical trials of complement inhibitors in IgAN.

Treatment Combinations

Combinations of immunosuppressants have been described for cIgAN presenting with nephrotic syndrome, as response to corticosteroids alone is poor.^{305,347} Improved responses have been reported with combinations such as azathioprine (2 mg/kg/day), mycophenolate mofetil (20–30 mg/kg/day), or cyclosporine, with or without glucocorticoids.^{305,347} These reports are, however, limited by small sample sizes and retrospective cohort designs.

MANAGING GLOMERULAR FIBROSIS

As in adults, there has been little progress in development of novel antifibrotic drugs for use in children, and in 2025 there are no antifibrotic drugs on the horizon for use in cIgAN.

IgA VASCULITIS IN ADULTS

INTRODUCTION

Immunoglobulin A vasculitis (IgAV) is a systemic, leukocytoclastic, small-vessel vasculitis. It was known as Henoch-Schönlein purpura (HSP) until the 2012 Chapel Hill International Consensus Conference for Nomenclature of Vasculitides when the term IgAV was introduced. IgAV is characterized by IgA1-dominant immune deposits involving skin, intestine, joints, and kidneys, causing a glomerular disease histologically indistinguishable from IgAN.³⁴⁸ When IgAV involves the kidneys, it is referred to as IgAV-nephritis (IgAV-N).

It was Heberden who provided the first clinical description of the syndrome in 1802. Schönlein described the association between purpura and arthritis in 1837, with Henoch later associating gastrointestinal and renal involvement.³⁴⁹

Compared with children, IgAV tends to be more severe in adults with more frequent kidney involvement and poorer longer-term outcomes. Kidney disease in IgAV presents more acutely than in IgAN. It has been suggested that IgAN and IgAV are part of a single spectrum of disease, as patients with IgAV may have persistent IgA immune complex-mediated glomerular disease after resolution of systemic vasculitis, while patients presenting with kidney-limited disease and diagnosed with IgAN have been reported to later develop the systemic manifestations of IgAV.³⁵⁰ There is also a large overlap in the immunopathologic mechanisms between the two conditions.³⁵¹ Similarly to IgAN, patients with IgAV-N can develop recurrence in kidney allografts.³⁵²

EPIDEMIOLOGY

IgAV occurs much less frequently in adults than in children, and the reported incidence varies greatly. This variation is in part due to a lack of standardized diagnostic criteria for adults and also depends on whether case enumeration is based on hospital or community-wide records. The annual incidence of adult IgAV based on hospital records has been reported to be 1.5 to 18 per million population.³⁵³⁻³⁵⁵ Using primary care data, the annual incidence of adult-onset IgAV in the United Kingdom is 22 per million population.³⁵⁶ An exceptionally high annual incidence for histologically proven IgAV of 50 per million population has been reported from a combined secondary/tertiary level center in Slovenia.³⁵⁷ Men are affected more frequently than women with a ratio of 1.5:1, and all ethnic groups are affected.³⁴⁹ Kidney involvement affects 45% to 85% of adults with IgAV compared with only 20% to 50% of children with IgAV.³⁵⁸

PATHOGENESIS

Similar to IgAN, the pathophysiology of IgAV-N follows the 4-hit hypothesis (see earlier). However, there are some key immunopathologic differences between IgAV and IgAN. Patients with IgAV have a more marked increase in circulating levels of IgE, compared with IgAN,³⁵¹ and increased levels of eosinophil cationic protein (ECP).³⁵⁹ In keeping with the pathophysiology of other small-vessel vasculitides, neutrophil extracellular traps (NETs) have been detected in the circulation of children with IgAV and in IgAV-affected tissue.³⁶⁰ It has been shown that activation of the IgA receptor Fc α RI (CD89) by IgA1 can mediate the process of NETosis in neutrophils,³⁶¹ and polymorphisms in CD89 associate significantly with the risk of IgAV (Kiryluk et al., unpublished) offering an explanation for differential effects of IgA immune complexes between IgAV and IgAN.

Circulating IgA immune complexes in IgAV are reported to be larger than those found in IgAN, reaching >19S by sucrose gradient ultracentrifugation.³⁶²⁻³⁶⁴ Compared with healthy subjects and IgAV patients without nephritis, patients with IgAV-N have significantly higher levels of circulating dg-IgA1³⁶³ and higher concentrations of IgA-sCD89. Circulating levels of IgA-IgG immune complexes are similar between patients with IgAV-N and IgAV without nephritis.³⁶⁵ Compared with IgAN patients, patients with IgAV-N have similar levels of dg-IgA1 and IgG anti-dg-IgA1 autoantibodies.³⁶⁶

Although purpuric skin lesions are not seen in IgAN, IgA deposition has been variably reported in skin biopsies from more than half of IgAN patients studied,^{367,368} lending further support to a spectrum of disease between IgAN and IgAV.

Acute upper respiratory infections are a recognized trigger of IgAV in children and adults, with *Streptococcal* spp. infection being the most frequent trigger. In older adults, malignancy is an important secondary cause of IgAV. The pathophysiologic link between malignancy and IgAV is unknown.

Clinical Relevance

In older adults, malignancy is an important secondary cause of IgA vasculitis occurring in 2% to 12% of adult cases. The link is unknown. It is recommended to perform age- and gender-appropriate cancer screening and to investigate any signs or symptoms suspicious of malignancy.

CLINICAL PRESENTATION

IgAV classically presents with four key symptoms (a tetrad), although not all need to be present: palpable purpura, arthritis/arthralgia, abdominal pain, and kidney disease. The characteristic rash consists of petechiae or palpable purpura and predominantly involves the lower extremities with extension to the upper extremities (Fig. 33.8A). Over time, and with severe disease, the lesions can become necrotic or hemorrhagic. Arthritis typically involves the large joints of the lower extremities and is usually symmetric but can migrate between joints. GI symptoms include colicky abdominal pain, with or without diarrhea, vomiting, or lower GI bleeding.

These symptoms develop over days to weeks and can appear in any order, although purpura and joint pains usually occur first, while kidney involvement typically manifests within a few days to 1 month after the onset of systemic symptoms. In one large series of 260 adults with IgAV in France, palpable purpura was present in all patients, with renal (70%), arthritis/arthralgias/myalgia (61%), and gastrointestinal involvement (51%) affecting the majority. About a third (30%) had an eGFR <60 mL/min/1.73 m² at presentation, and 32% had persistent kidney impairment after a median follow-up of 15 years.³⁶⁹ Upper respiratory tract infection is a common predisposing factor, present in 34% of adults in one Turkish cohort.³⁷⁰ Recent history of infection at time of presentation is more common in adults younger than 30 years old, and arthritis is a less common feature of IgAV in adults older than 60 years old. The proportion of patients found to have necrotic purpura increases with age at presentation.³⁷¹

Kidney involvement most commonly presents with nonvisible or visible hematuria and mild or moderate proteinuria: Nephrotic syndrome or severe kidney dysfunction affects a minority. Patients presenting at an older age appear to have more severe kidney disease, with a higher likelihood of kidney involvement at diagnosis and higher risk for subsequent CKD development.³⁷² Among 161 patients with IgAV included in the Cure Glomerulonephropathy prospective cohort study, adults had a significantly lower median eGFR at the time of kidney biopsy compared with children (67 vs. 109 mL/min/1.73 m²) and had a significantly higher risk for development of advanced CKD (eGFR <30 mL/min/1.73²) during follow-up (20% vs. 4%).³⁷³ In a Turkish study, hypertension, hematuria, proteinuria, and renal insufficiency at diagnosis, as well as CKD during follow-up, were also more frequent in adults.³⁷⁰ In a large North American cohort, ulcerative skin lesions and nephrotic-range proteinuria were more common in adults while gastrointestinal and joint involvement were more common in children.³⁷⁴

Beyond the classic tetrad of symptoms, involvement of other organs has more rarely been described. Lung involvement can occur in a small number of patients (2.4% in one U.S. cohort), manifesting as diffuse pulmonary alveolar hemorrhage or usual interstitial pneumonia or interstitial fibrosis.³⁷⁵ Central nervous system involvement,³⁷⁶ episcleritis/keratitis,³⁷⁷ myocarditis,³⁷⁸ and epididymo-orchitis³⁷⁹ are extremely rare in adults but have been reported.

Malignancy is associated with IgAV in 2% to 12% of adult cases, about two-thirds of which are solid organs, and one-third are hematologic malignancies.³⁸⁰ Malignancy-associated IgAV more commonly affects older males. Solid organ malignancies are typically present before diagnosis of IgAV, most commonly affecting mucosal organs, namely lung, colon, or bladder.³⁸¹ Hematologic malignancies tend to be diagnosed synchronously or after the diagnosis of IgAV.³⁸⁰ Malignancy-associated IgAV presents more frequently with necrotic purpura and pulmonary alveolar hemorrhage than IgAV without associated malignancy.³⁸¹ It is recommended to perform age- and gender-appropriate cancer screening in patients presenting with IgAV and to investigate for any signs or symptoms suspicious of malignancy.¹⁴⁰ TNF inhibitors have been identified as a drug-trigger for IgAV.^{382,383}

IgAV-N can recur after kidney transplantation and has been reported to result in graft loss in about 10% of cases.³⁸⁴

DIAGNOSIS OF IgAV

The most recent diagnostic criteria for IgAV come from The European League Against Rheumatism, Pediatric

Rheumatology International Trials Organization, and Pediatric Rheumatology European Society (EULAR/PRINTO/PRES) as outlined in Table 33.7.³⁸⁵ Although these criteria were defined for pediatric disease, they have 99% sensitivity and 86% specificity in adults,³⁸⁶ performing better than the original American College of Rheumatology criteria.

In the correct clinical context, where other causes of cutaneous purpura have been excluded, urinalysis showing hematuria and proteinuria may be sufficient to confirm a diagnosis of IgAV-N. Kidney biopsy should be performed when there is rapidly progressive glomerulonephritis, persistent nephritis with proteinuria >1 g/day, or persistent impaired kidney function.¹⁴⁰ Kidney biopsy should also be considered if there is clinical uncertainty regarding diagnosis of the systemic disease or when it is not clear if abnormal urinalysis is due to IgAV-N or an alternative concomitant glomerular disease. Kidney tissue can be stained for IgA deposition as part of the EULAR/PRINTO/PRES diagnostic criteria for IgAV and allows characterization of the associated kidney disease.

DIFFERENTIAL DIAGNOSIS

The differential diagnosis in adults includes other systemic diseases presenting with vasculitis including bacteremia, hypersensitivity vasculitis, ANCA-associated vasculitis, cryoglobulinemic vasculitis, systemic lupus erythematosus, and infection-related glomerulonephritis (especially the IgA-dominant form). Other diagnostic considerations include petechiae or purpura due to clotting disorders or thrombocytopenia (e.g., TTP and disseminated intravascular coagulation).

Table 33.7 Sensitivity and Specificity of Different Diagnostic Criteria for IgAV in a Pediatric Population, used to Define the Ankara 2008 Classification^{384a}

Final EULAR/PRINTO/PRES HSP Criteria (with Glossary) and Classification Definition (Sample 973)				
Criterion	Glossary	Sensitivity (%)	Specificity (%)	AUC (%)
Purpura (mandatory criterion)	Purpura (commonly palpable and in crops) or petechiae, with lower limb predominance, ^a not related to thrombocytopenia	89	86	87.5
1. Abdominal pain	Diffuse abdominal colicky pain with acute onset assessed by history and physical examination. May include intussusception and gastrointestinal bleeding	61	64	62.2
2. Histopathology	Typically leukocytoclastic vasculitis with predominant IgA deposit or proliferative glomerulonephritis with predominant IgA deposit	93	89	91.1
3. Arthritis or arthralgias	Arthritis of acute onset defined as joint swelling or joint pain with limitation on motion Arthralgia of acute onset defined as joint pain without joint swelling or limitation on motion	78	42	59.9
4. Renal involvement	Proteinuria >0.3 g/24 h or >30 mmol/mg of urine albumin/creatinine ratio on a spot morning sample Hematuria or red blood cell casts: >5 red blood cells/high power field or red blood cells casts in the urinary sediment or ≥2+ on dipstick	33	70	51.4
HSP EULAR/PRINTO/PRES Ankara 2008 classification definition: κ 0.90 (95% CI 0.84–0.96)	Purpura or petechiae (mandatory) with lower limb predominance ^a and at least 1 of the 4 following criteria: Abdominal pain Histopathology Arthritis or arthralgia Renal involvement	100	87	93.5

^aFor purpura with atypical distribution a demonstration of an IgA deposit in a biopsy is required.

AUC, Area under the curve; EULAR, European League Against Rheumatism; HSP, Henoch--Schönlein purpura; PRES, Paediatric

HISTOPATHOLOGY

While it is widely accepted that the kidney histopathology of IgAV-N is indistinguishable from that of IgAN, some subtle differences have been described. Patients with IgAV-N have more prominent capillary wall deposition of IgA (Fig. 33.7), likely due to the larger size of IgA immune complexes. The λ chain predominance described in IgAN is not seen in IgAV. Inflammatory lesions such as endocapillary proliferation, mesangial proliferation, and crescents are seen more frequently in IgAV. Glomerular deposition of fibrin is more common in IgAN-V than IgAN.³⁵¹

A skin biopsy can be considered to diagnose IgAV when there is a suspicion of IgAV-N, but the risk of kidney biopsy is

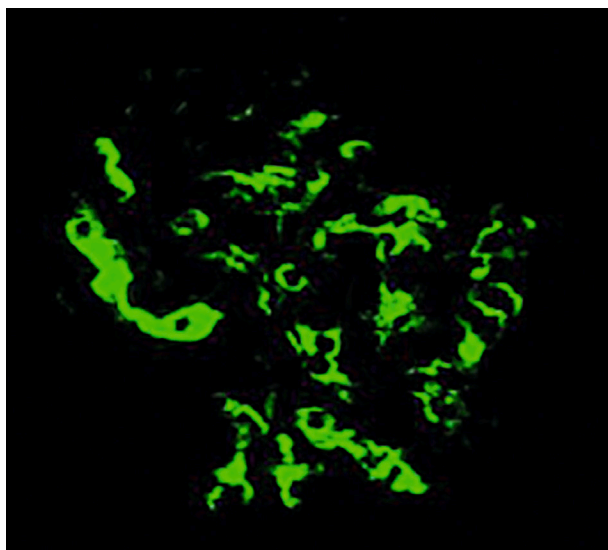


Fig. 33.7 Pericapillary IgA deposition in glomerulus from patient with IgAV.¹⁷⁰

felt to be unacceptable, and alternative diagnoses have been excluded. Diagnostic yield is, however, highly variable. A predominantly leukocytoclastic vasculitis affecting the small superficial vessels is typically seen, although other phases of the inflammatory process with lymphocytic vasculitis and perivascular inflammation may also be seen.³⁸⁷ When leukocytoclastic vasculitis (LCV) has been confirmed, vascular deposition of IgA has a reported sensitivity of up to 81% and specificity of up to 83% for IgAV.³⁸⁸ Timing and location of the skin biopsy are important. Perilesional biopsies are more likely to contain IgA deposits than necrotic areas, and there is higher likelihood of an absence of IgA staining with increasing age of the lesion. Ideally, established nonnecrotic lesions should be biopsied to look for LCV by LM, while tissue taken from new perilesional sites is most likely to demonstrate IgA deposition by direct immunofluorescence³⁸⁷ (Fig. 33.8B).

RISK PREDICTION

Greater amounts of proteinuria and lower eGFR are significantly associated with worse renal outcomes in IgAV.^{371,390}

The International Study of Kidney Disease in Children (ISKDC) classification was developed for pediatric disease³⁹¹ and was not found to be useful in adults³⁹²; however, an adaptation of this histologic score by Pillebout and colleagues³⁷¹ did significantly predict renal outcome in adults. The ISKDC histologic scores have been criticized for not emphasizing important histologic lesions of chronicity such as tubulointerstitial fibrosis and glomerulosclerosis.^{390,393} More recently, the MEST-C score developed for IgAN risk prediction has been shown to also have prognostic utility in IgAV. IgAV patients with endocapillary proliferation (E1) are more likely to develop renal failure over time regardless of immunosuppression.³⁹⁰

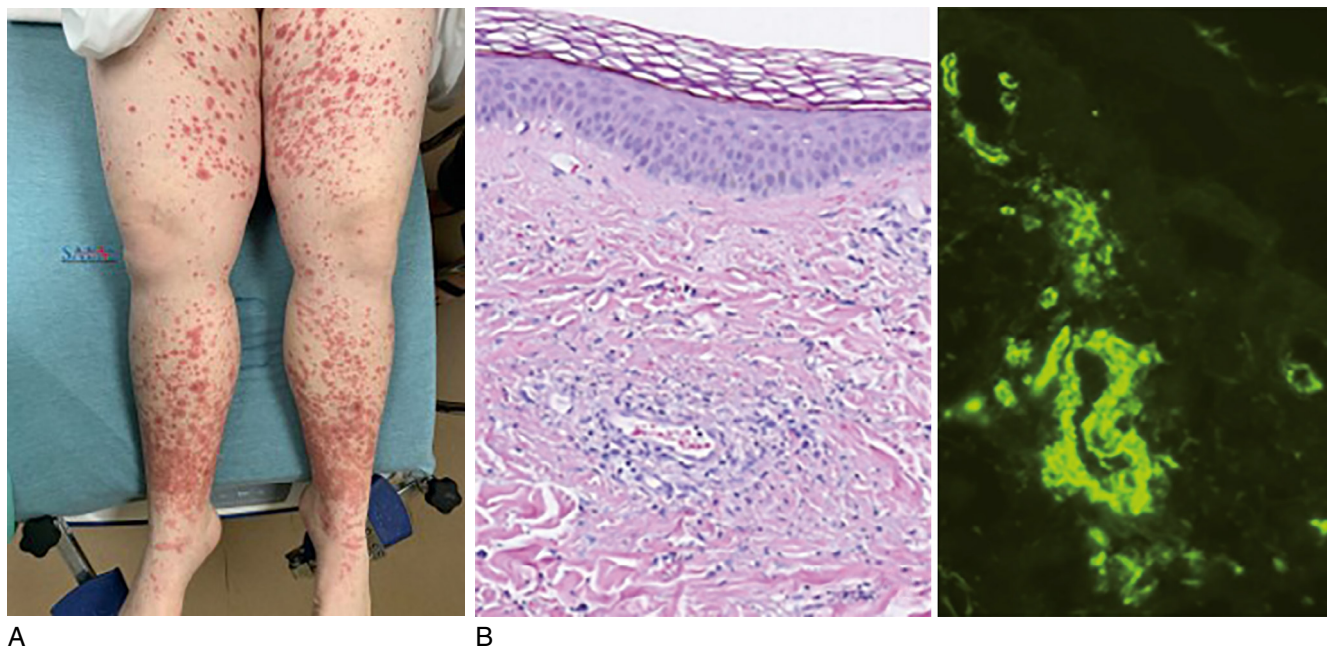


Fig. 33.8 (A) Purpuric rash in IgAV. (B) Leukocytoclastic rash with IgA deposition skin biopsy.³⁸⁹ (A, Used with the permission of Dr. Erin Shufflebarger, author, and Megan Schagrin, CEO at SAEM.)

PROGNOSIS

Overall, IgAV (with or without nephritis) resolves spontaneously in up to 89% of adults.³⁹⁴ Considering only patients with confirmed IgAV-N, the risk of progression to CKD is higher in adults than in children, estimated to be between 35% and 70% beyond 5 years of follow-up.³⁹³ Progression to kidney failure is seen in about 10% over 15 years.³⁷¹ Interestingly, adult IgAV-N patients treated with immunosuppression have been shown to fall into one of two clinical phenotypes either with 1. stable kidney function over time (80%) or 2. initial improvement in kidney function, likely reflecting glomerular hyperfiltration, followed by progressive decline in eGFR (20%).³⁹⁰ Mortality in IgAV-N patients after a 14.8-year median follow-up was 26% in one study. Cancer is an important cause of death in these patients.³⁷¹ Among patients with severe IgAV included in the CESAR study, the overall mortality rate at 1-year follow-up was 13%.³⁹⁵

TREATMENT OF IgAV AND IgAV-NEPHRITIS IN ADULTS

Multiple studies in children with systemic IgAV have shown that early use of corticosteroids does not prevent development of nephritis,^{396,397} and this approach is therefore not recommended.¹⁴⁰ As with IgAN, when IgAV causes nephritis, treatment should focus on the predominant drivers of nephron loss.

MANAGING THE GENERIC RESPONSES TO IgA VASCULITIS NEPHRITIS-INDUCED NEPHRON LOSS

Treatment approaches incorporating lifestyle modification and drugs to minimize systemic and glomerular hypertension should be employed while simultaneously addressing cardiovascular risk factors present at the time of diagnosis or that become apparent during follow-up (as described earlier for IgAN). It is important to acknowledge that patients with IgAV-N were excluded from the PROTECT trial of sparsentan in IgAN,³⁹⁸ and it is not clear how many patients with IgAV-N were included in the trials of SGLT2i in nondiabetic kidney disease.²⁴¹ In the setting of an RPGN, commencement of RAASi, SGLT2i, and endothelin receptor antagonists should be avoided until kidney function has been stabilized by therapies targeting the immunological aspects of the disease.

MANAGING THE IgA VASCULITIS-SPECIFIC DRIVERS FOR NEPHRON LOSS

As with IgAN, the key initiators of nephron loss in IgAV-N are 1. the production of pathogenic IgA and formation of IgA immune complexes, 2. the inflammatory, and 3. the profibrotic glomerular responses to glomerular IgA immune complex accumulation. A similar treatment approach should therefore be adopted in IgAV-N. In comparison with IgAN, there have been no phase 2 or 3 trials of new therapies in IgAV-N and available data are limited to retrospective cohort studies, most older than 10 years old. The threshold for intervention beyond optimized supportive care is unclear outside of a presentation with an RPGN¹⁴⁰; current guidelines recommend considering immunomodulatory

therapy if proteinuria remains >1 g/day despite 3 months of maximum supportive care.¹⁴⁰

Blocking the Production of Pathogenic IgA

B cell/plasma cell depletion. While there have been no randomized controlled trials of rituximab in IgAV-N, an observational study supports its potential therapeutic role in the treatment of refractory IgAV-N.³⁹⁹ Current small phase 2 trials of CD38+ plasma cell depletion in IgAN have allowed inclusion of patients with IgAV-N and results are awaited (see Table 33.6).

B cell/plasma cell modulation. Patients with IgAV-N have been excluded from trials of Nefecon and drugs targeting BAFF/APRIL, so it is not possible to comment on the role of these therapeutic approaches in IgAV-N.

In life-threatening diseases, intravenous immunoglobulin has been reported in single cases and small retrospective cohorts to improve outcomes.⁴⁰⁰

Inhibiting proliferation and signaling. There is limited evidence for a benefit of mycophenolate mofetil (with reduced-dose corticosteroids)⁴⁰¹ and cyclosporin A⁴⁰² in IgAV-N; however, these data are from single observational studies and should be interpreted cautiously.

For patients with rapidly progressive IgAV-N, the KDIGO 2021 guidelines recommend treating the same way as for AAV with cyclophosphamide and corticosteroids.¹⁴⁰ The CESAR randomized controlled trial in adults with severe IgAV (including IgAV-N with endocapillary or extra capillary proliferation, severe gastrointestinal involvement, pulmonary hemorrhage, cardiac involvement, central nervous system involvement, and/or episcleritis) compared the utility of intravenous cyclophosphamide and corticosteroids compared with corticosteroids alone. Only 54 of the 200 target patients were recruited, and among these, no statistically significant differences were seen between groups.³⁹⁵

Plasma exchange. In life-threatening diseases, case reports suggest potential therapeutic benefits with plasma exchange,⁴⁰³⁻⁴⁰⁵ consistent with reported benefits in severe ANCA-associated vasculitis.^{406,407}

Managing Glomerular Inflammation

Systemic glucocorticoids. Consistent with many other inflammatory glomerular diseases, systemic glucocorticoid therapy has been used in adult IgAV-N for many years.³⁷¹ However, there have been no trials of glucocorticoid monotherapy in IgAV-N. Glucocorticoid-related toxicity and poor tolerability have been significant concerns for both clinicians and patients.⁴⁰⁸ Outside of patients presenting with RPGN, there is little evidence for long-term effectiveness of systemic glucocorticoids in IgAV-N.

Complement inhibition. Patients with IgAV-N have been excluded from the trials of complement inhibitors in IgAN, and while there is convincing evidence for complement activation in IgAV-N, it is unlikely that there will be significant clinical trial data available in the near future to support their use in IgAV-N.

IgA VASCULITIS IN CHILDREN

INTRODUCTION

IgA Vasculitis (IgAV), previously known as Henoch-Schönlein purpura (HSP), shares clinical, histologic, and pathophysiologic similarities with IgAN.⁴⁰⁹⁻⁴¹² IgAV is a multisystem vasculitic disease characterized by palpable nonthrombocytopenic purpura of the skin, abdominal pain, arthritis, and kidney involvement, with characteristic histology marked by leukocytoclastic vasculitis of the skin and proliferative glomerulonephritis with predominance of IgA deposition.⁴¹³

From a kidney point of view, the evolution of IgAV is highly variable. Kidney involvement is reported in 20% to 54% of children with IgAV and is referred to as IgA vasculitis with nephritis (IgAV-N); risk factors for kidney involvement are not well established.⁴¹⁴

Consistent with a disease spectrum between IgAN and IgAV-N, some cases of IgAV-N will evolve into a paradigm of recurrent relapses of kidney disease without clinical purpura, mirroring childhood IgAN—this presentation carries a worse prognosis.^{415,416}

EPIDEMIOLOGY

IgAV is a common condition in childhood, with an estimated annual incidence between 3 and 26.7 cases/100,000 children, surpassing the incidence in adults 2- to 33-fold.^{417,418} The reported frequency of IgAV in children has been slowly increasing, which may reflect changes in the criteria used to identify cases in epidemiologic studies.^{419,420} The true incidence rate may be higher, as children with limited cutaneous IgAV, which does not necessitate hospitalization, may not be captured in health care records.

There are ethnic variations in IgAV incidence, with rates being three to four times higher in Caucasian and Asian children compared with their counterparts of African ethnicity.⁴²⁰ Reported annual incidence rates vary between countries, ranging from 6.79 per 100,000 children in Croatia, 18 per 100,000 children in France,⁹⁹ to 55.9⁴²¹ per 100,000 children in Korea.⁴²²

Approximately 75% to 90% of IgAV cases occur in children younger than the age of 10, with the highest incidence observed between 4 and 7 years of age, where incidence is reported to be 70.3 cases per 100,000 individuals.⁴²⁰ This may be due to the heightened susceptibility of children in this age bracket to certain infectious pathogens.

As in adulthood, IgAV in children is more commonly seen in males with a prevalence ratio of approximately 1.5:1 for males to females.⁴²⁰

With regard to kidney involvement, IgAV-N is more commonly seen in adults, reported to affect 45% to 85% of IgAV cases, in comparison with 20% to 50% of IgAV cases in children.³⁵⁸

PATHOPHYSIOLOGY

Although similarities between the pathophysiology of IgAV-N and IgAN have been reported, there are some significant differences.⁴²³ While some studies have shown increased

dg-IgA1 levels in children with IgAV, this has not been uniformly reported, with some studies showing no change in dg-IgA1 levels in the serum and mesangial IgA deposits.⁴²⁴ Unlike in IgAN, a significant role for neutrophil activation and neutrophil extracellular trap-mediated endothelial cell injury has been proposed to explain the multisystem organ injury seen in IgAV.⁴²⁵

Extensive evidence supports the relationship between infection and the onset of IgAV in childhood. IgAV presentations are typically most frequent during the autumn and winter seasons, with lower rates observed in the summer months,⁴¹⁹ aligning with the seasonal pattern of many upper respiratory infections.⁴²⁶ Up to 70% of children with IgAV present having had a prodromal infection or with symptoms of infection at the time of presentation, with respiratory tract infections being the most prevalent, followed by gastrointestinal infections, cellulitis, and urinary tract infections. Common infectious pathogens associated with childhood IgAV in China include *Streptococci* spp., *Helicobacter pylori*, *Mycoplasma pneumoniae*, human parvovirus B19, and hepatitis A virus.⁴²⁷ Relapses of IgAV have been linked to reinfection with specific pathogens. Of interest, treatment of *H. pylori* infection has been shown to result in IgAV remission, with *H. pylori* reinfection associated with IgAV relapse.^{428,429}

Pathogen-induced activation of the mucosal immune system and mucosal inflammation have been associated with a number of potential triggers for the development of IgAV: Mucosal-type IgA is prone to form immune complexes in the circulation and deposit in the kidney; pathogens with similar antigenic structures to vessel walls have been suggested to drive the formation of IgA antiendothelial cell antibodies;⁴³⁰ infections can also raise levels of proinflammatory factors such as IL-6 and PRRs like TLR9, alongside the B cell survival factor APRIL, all of which promote the synthesis of dg-IgA1.⁴³¹

CLINICAL PRESENTATION

Two-thirds of children with IgAV will not have abnormal urinalysis. Among children with IgAV who develop IgAV-N (one-third), hematuria and/or proteinuria is seen within 2 to 4 weeks for 50% to 80% of cases and, for the majority, these manifestations will be minor and self-limiting.⁴³²

IgAV-N can present with acute kidney injury, particularly in the acute phase of the disease.⁴¹⁷ This may be due to an RPGN, severe hematuria resulting in tubular injury or as a consequence of potentially life-threatening complications of IgAV, such as severe gastrointestinal bleeding, intestinal intussusception, or intestinal perforation.⁴³³

DIAGNOSIS

Previously, the diagnosis of IgAV relied on the 1990 American College of Rheumatology classification criteria,⁴³⁴ which lacked specificity for children, limiting its applicability. In contrast, the 2008 criteria established by the European League against Rheumatism/Pediatric Rheumatology International Trials Organization/Pediatric Rheumatology European Society (EULAR/PRINTO/PRES) provided specific diagnostic guidelines for children,

mandating the presence of purpura or petechiae, along with at least one additional criterion such as abdominal pain, arthritis, kidney involvement, or biopsy-proven IgA deposition at any site (sensitivity 100%, specificity 87%).³⁸⁵ These criteria are now routinely used by pediatricians (see Table 33.7).

It is imperative that all children suspected of having IgAV undergo proactive evaluation for kidney involvement at the initial diagnosis of IgAV. This evaluation should include blood pressure measurement; a thorough clinical assessment for peripheral edema; first-morning void urinalysis, with quantification of proteinuria when urinalysis shows proteinuria; and measurement of kidney function.⁴³⁵ Urinalysis should be conducted at first presentation and repeated at least monthly for a minimum of 6 months.

Kidney biopsy is required to make a definitive diagnosis of IgAV-N. There are, however, few data to help guide when to perform a kidney biopsy in children with IgAV. Most pediatric nephrologists will follow the SHARE guidelines⁴³⁵ and perform a kidney biopsy in children with IgAV who exhibit persistent proteinuria, nephrotic-range proteinuria, and/or reduced eGFR.

HISTOPATHOLOGY

In practical terms, the histopathological features of IgAV-N are considered to be the same as those for IgAN. The ISKDC classification is still the most frequently used for histologic analysis of renal biopsy findings in IgAV-N with 6 grades (Table 33.8).⁴³⁶⁻⁴³⁸ Nevertheless, many treatment centers are now using the Oxford classification, especially since endocapillary hypercellularity (E1), a factor not accounted for in the ISKDC classification, has been shown to be a significant predictor of late eGFR decline.

RISK PREDICTION IN IgAV

The long-term prognosis in IgAV is heavily influenced by kidney involvement, making early diagnosis and timely treatment crucial for managing IgAV in children. Two key clinical features, age, and duration of rash, have been identified as predictors of IgAV-N.⁴³⁹⁻⁴⁴¹ In a study involving 535 patients, older age at disease onset, purpura on the upper limbs or face, and occult blood in the stool predicted an increased risk of developing IgAV-N during follow-up.⁴⁴⁰ In addition, Buscatti and colleagues⁴⁴¹ found that the persistence of rash for more than 1 month was a significant predictor of IgAV-N development. A significantly elevated level of D-dimer has been reported to indicate a state of hyperfibrinolysis in the early stages of IgAV. Of interest, reports have also suggested that decreased serum level of IgG and increased levels of D-dimer are predictive of development of IgAV-N in patients with IgAV.⁴³⁹

In a review of 12 studies involving 1133 children, permanent renal impairment was not observed in any patients with normal urinalysis. It occurred in 1.6% of individuals with isolated urinary abnormalities and in

Table 33.8 ISKDC Histopathologic Classification of IgAV-N^{435a,435b}

ISKDC Grade	Description
Grade I	Minimal alterations
Grade II	Mesangial proliferation
Grade III	Proliferation or sclerosis with <50% crescents ([a] focal or [b] diffuse)
Grade IV	Mesangial proliferation or sclerosis with 50%–75% crescents ([a] focal or [b] diffuse)
Grade V	Mesangial proliferation or sclerosis with >75% crescents ([a] focal or [b] diffuse)
Grade VI	Membranoproliferative like glomerules

19.5% of those who progressed to nephritic or nephrotic syndrome.⁴³²

Kidney failure risk prediction in IgAV is poorly understood, and it is not known whether the International IgAN Prediction Tool for children can be applied to children with IgAV-N, even though the kidney biopsy features of the two diseases are almost indistinguishable.⁴³⁷ Evaluation of the extent of kidney damage in IgAV-N has, until recently, been less well defined than in IgAN. The ISKDC classification primarily emphasizes active inflammation, which is commonly seen in IgAV-N in children, while overlooking vascular and tubulointerstitial changes, and has been widely used in studies of IgAV-N in childhood.⁴³⁶⁻⁴³⁸ As with adults, there has been an increasing recognition of the importance of glomerulosclerosis and interstitial fibrosis in predicting long-term outcomes in childhood.^{442,443} The Oxford classification is becoming widely adopted in pediatric practice, and in an international study involving 361 patients (262 children, 99 adults), endocapillary hypercellularity (E1), a factor not accounted for in the ISKDC classification, was shown to be a significant predictor of late eGFR decline. Endocapillary hypercellularity was also shown to be more predictive of outcome than the presence of crescents, which serve as the main differentiator among ISKDC grades III, IV, and V. Furthermore, MEST-C scores have been shown to correlate with eGFR and extent of proteinuria at the time of biopsy.³⁹⁰

When comparing outcomes between IgAN and IgAV-N in children, most studies report worse long-term outcomes in IgAN.⁴³⁶ IgAN tends to be more common in children older than 10 years of age with 30% to 40% of cases progressing to kidney failure within 10 to 20 years of initial presentation,⁴⁴⁴ whereas many cases of IgAV-N display modest urine abnormalities, which may fluctuate with a tendency to resolve spontaneously.^{423,435} In two separate studies, children with IgAV and IgAN were compared, revealing distinct clinical profiles and outcomes. In a Spanish study that included 142 IgAV patients and 61 IgAN patients of all ages, including children, the IgAV patients were younger, had lower rates of impaired eGFR (25% in IgAV vs. 63.4% in IgAN) and nephrotic syndrome (12.5% vs.

43.7%), and better long-term outcomes including reduced rates of dialysis (2.9% in IgAV vs. 43.5% in IgAN), renal transplant (0% vs. 36%), and chronic renal insufficiency (4.9% vs. 63.8%) after a median follow-up of 11 years.⁴³⁸ In a Chinese study comparing 31 cIgAN patients with 120 children with IgAV-N, cIgAN patients were significantly older and there were more sclerotic changes in cIgAN kidney biopsy samples (35.5% vs. 3.1%) compared with more endocapillary proliferation in IgAV-N kidney biopsy samples (65.6% vs. 29%).⁴³⁶ During a 34-month follow-up, IgAV-N children demonstrated a more favorable prognosis, with 72.5% achieving complete remission compared with only 19.4% in the cIgAN group.

TREATMENT

MANAGING THE GENERIC RESPONSES TO IgAV-N INDUCED NEPHRON LOSS

Optimized Supportive Care

RASB with ACE inhibitors or ARBs have been extensively used as proteinuria-lowering therapies in children with IgAV-N, but there are no prospective studies to substantiate their use or to provide guidance on the timing of initiation. Their use is justifiably based on similarities with cIgAN.^{140,331}

MANAGING THE IgAV-N-SPECIFIC DRIVERS FOR NEPHRON LOSS

Blocking the Production of Pathogenic IgA

Inhibiting lymphocyte proliferation and signaling

Mycophenolate mofetil. Mycophenolate mofetil (MMF) has been examined in single-center studies. The largest single-center study ($n = 61$) found that adding MMF (20–30 mg/kg/day) to glucocorticoids in patients with IgAV-N and nephrotic-range proteinuria resulted in a higher rate of complete remission at 6 months (26 of 41 patients) compared with glucocorticoids alone (7 of 20 patients). However, at 23 months, both groups exhibited similar rates of complete remission.⁴⁴⁵ Other smaller, single-center observational studies have also reported a similar favorable response to MMF, although the duration of MMF therapy has varied.^{446,447}

Cyclophosphamide. There are few studies of cyclophosphamide treatment in IgAV-N in childhood. In one study by Tarshish and colleagues,⁴⁴⁸ 56 children with IgAV-N and crescents on kidney biopsy were given either oral cyclophosphamide (at 90 mg/m²/day for 42 days) or no treatment. After 7 years of follow-up, there was no significant difference in kidney outcomes between the two groups, including persistent kidney disease (based on proteinuria and eGFR) and kidney failure.⁴⁴⁸

Calcineurin inhibitors. In a small, nonrandomized, nonblinded cohort study in children with biopsy-proven IgAV-N, classified as ISKDC grade III or IV, or grade II with nephrotic syndrome, Jauhola and colleagues⁴⁴⁹ compared the effect of cyclosporine to methylprednisolone pulses. All children treated with cyclosporine achieved remission of nephrotic-range proteinuria at 3 months, whereas only 50% of those treated with prednisolone achieved remission.

Azathioprine and mizoribine. In a study of 19 children with IgAV-N and serum albumin levels ≤ 2.5 g/dL and ISKDC classification grade IV or higher, Ninchoji and colleagues⁴⁵⁰ evaluated the effectiveness of combination therapy with glucocorticoids, azathioprine or mizoribine, warfarin, and dipyridamole. Mizoribine, an imidazole nucleoside, exerts its effects through its metabolite, MZ-5-P, which selectively inhibits inosine monophosphate synthetase and guanosine monophosphate synthetase. All patients experienced disappearance of proteinuria without kidney dysfunction and no serious side effects.⁴⁵⁰ Some 10 years later, Nagai and colleagues⁴⁵¹ reported favorable outcomes in a small number of children with IgAVN treated with a combination of glucocorticoids, mizoribine, and RAS inhibitors. It must be noted that both studies were small, nonrandomized, and nonblinded.

B cell/plasma cell depletion. In a retrospective study of 8 children with IgAV and persistent refractory disease treated with rituximab, CD20⁺ depletion resulted in fewer hospitalizations and an increased likelihood of steroid discontinuation.³³⁵ Seven of eight children met remission criteria, defined as no active rash, arthritis, nephritis (hematuria and proteinuria), or gastrointestinal disease following rituximab.

Managing Glomerular Inflammation

Systemic glucocorticoids. Early studies suggested potential benefits of corticosteroids in IgAV, but their small size and poor design limit their reliability. Recent meta-analyses⁴⁵² and randomized controlled trials⁴⁵³ have failed to conclusively demonstrate corticosteroid efficacy; however, corticosteroids are commonly used to treat IgAV-N in children.

There are multiple glucocorticoid protocols in use. Many involve methylprednisolone pulses followed by oral glucocorticoids in gradually decreasing doses. This approach is commonly used in cases where kidney histology reveals active proliferative lesions in $>50\%$ of glomeruli (ISKDC $\geq$ IV) or in instances of nephrotic proteinuria or rapidly progressive glomerulonephritis (RPGN).⁴⁵⁴

In one prospective, uncontrolled study, 38 children with nephrotic syndrome and/or crescents affecting $>50\%$ of glomeruli were treated with three intravenous methylprednisolone pulses followed by oral prednisone for 3 months.⁴⁵⁵ Four patients progressed to kidney failure, with three of those four having received treatment late in the course of their disease. This study suggests that early therapy may be beneficial in preventing irreversible glomerular injury. In a retrospective study involving 186 children with IgAV-N, children with nonnephrotic proteinuria were significantly more likely to respond to oral glucocorticoid therapy compared with supportive care alone (resulting in the disappearance of isolated hematuria or reduction of proteinuria). Children with nephrotic-range proteinuria were significantly more likely to respond to methylprednisolone pulses than oral glucocorticoid therapy.⁴⁵⁶

Current SHARE Treatment Recommendations for IgAV-N

SHARE (Single Hub and Access point for pediatric Rheumatology in Europe) is a working-group of experts in pediatric rheumatology and nephrology. They provide internationally agreed consensus recommendations for diagnosis and treatment for children with IgAVN. The SHARE initiative recommends corticosteroids as the primary treatment for IgAVN in children, regardless of severity.⁴³⁵ For severe cases, the SHARE initiative suggests a combination of intravenous cyclophosphamide and high-dose steroid therapy to induce remission, followed by maintenance therapy with azathioprine or MMF in combination with low-dose steroids.⁴³⁵

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